

群论及其应用

Week 1

Part 1 Review and study basic concepts & basic theory by studying important classes of gps.

Part 2 Group actions

Permutation groups — O' Nan-Scott thm

Part 3 Linear gps — classical groups & subgroups.

Cyclic groups (denoted by C_n)

$G = \langle g \rangle = \{1, g, g^2, g^3, \dots\}$ (or $\{0, 1, 2, \dots\}$ sometimes written additively since commutative)

- Cyclic groups are uniquely determined by the orders.
- subgroups are cyclic
- factor groups are cyclic

★ $\text{Aut}(C_n)$ ($C_n = \langle g \rangle$)

prop $\text{Aut}(C_n) \cong (\mathbb{Z}/n\mathbb{Z})^\times = \{\bar{x} \mid (x, n) = 1\} = U(n).$

pf: $\forall \phi \in \text{Aut}(C_n)$, $\phi(g) = g^i$ for some $i \in \mathbb{Z}$ and the integer i uniquely determines ϕ . Denote this automorphism by ϕ_i . We have $(i, n) = 1$ since g^i must have the same order as g . On the other hand, $\forall a$ s.t. $(a, n) = 1$, the map $g \mapsto g^a$ is an automorphism of C_n . Hence we have a surjective map

$$\psi: \text{Aut}(C_n) \rightarrow (\mathbb{Z}/n\mathbb{Z})^\times$$

$$\phi_a \mapsto a \pmod{n}$$

ψ is obviously a homo and injective.

▷ Aut $(C_p) = C_{p-1} = \langle 6 \rangle$
 use the fact that $\mathbb{F}_p^\times$ is cyclic

Recall let $F = \mathbb{G}_F(p^2)$
 $F^\times = C_p^\times = C_p \oplus \dots \oplus C_p$
 $F^\times = C_{p^e} - 1$

▷ Aut $(C_{p^e}) = C_{p^{e-1}(p-1)}$ where $e \geq 2$, p is odd prime.
 pf: $r = \lambda$ or $p + \lambda$ is a primitive root for C_{p^e} where $\text{order}_p(\lambda) = p-1$
 $(p+\lambda, p) = 1 \Rightarrow (p+\lambda, p^e) = 1 \Rightarrow (p+\lambda)^{p^{e-1}(p-1)} \equiv 1 \pmod{p^e}$

It suffices to show $p^{e-1}(p-1)$ is the smallest integer n s.t.
 $(p+\lambda)^n \equiv 1 \pmod{p^e}$ or $\lambda^n \equiv 1 \pmod{p^e}$.

First note that $p-1 \mid \text{order}_{p^e}(r)$ since $r^{p-1} \equiv 1 \pmod{p^e}$
 yields $r^{p-1} \equiv 1 \pmod{p}$ so $p-1 \mid |r|$. Then it
 suffices to show $(r)^{p^{e-1}(p-1)} \not\equiv 1 \pmod{p^e}$.

When $e=2$, $(p+\lambda)^{p-1} \equiv \lambda^{p-1} + p(p-1)\lambda^{p-2} \pmod{p^2}$

Here we use a lemma: $\forall \lambda$ primitive root of C_p , either λ
 or $p+\lambda$ is a primitive root for $(\mathbb{Z}/p^2\mathbb{Z})^\times$.

By above, $(p-1) \mid |\lambda|$, so $|\lambda| = p-1$ or $p(p-1)$. If $|\lambda| = p-1$,
 then $(p+\lambda)^{p-1} \equiv \lambda^{p-1} + p(p-1)\lambda^{p-2} \pmod{p^2} \equiv p(p-1)\lambda^{p-2} \pmod{p^2}$
 Then since $p \nmid \lambda^{p-2}$, $(p+\lambda)^{p-1} \not\equiv 1 \pmod{p^2}$, so $p+\lambda$ is
 a primitive root for $(\mathbb{Z}/p^2\mathbb{Z})^\times$.

Suppose it works for $e \leq n$, i.e. $r^{p^{n-1}(p-1)} \not\equiv 1 \pmod{p^n}$.

Since $r^{p^{n-2}(p-1)} \equiv 1 \pmod{p^{n-1}}$, we have $r^{p^{n-1}(p-1)} = 1 + dp^{n-1}$
 for some $(d, p) = 1$. Then $r^{p^{n-1}(p-1)} = (1 + dp^{n-1})^p \equiv$
 $1 + dp^n + \binom{p}{2} d^2 p^{2n-2} \pmod{p^{n+1}} \equiv 1 + dp^n \not\equiv 1 \pmod{p^{n+1}}$

Remark: We actually proved that if r is a primitive root for
 $(\mathbb{Z}/p^2\mathbb{Z})^\times$, then r is a primitive root for $(\mathbb{Z}/p^k\mathbb{Z})^\times$ for any k .

And we prove existence of such r by noting that if λ is a
 primitive root for C_p , then λ or $p+\lambda$ is a primitive root of $(\mathbb{Z}/p^2\mathbb{Z})^\times$.

pf 2: prove that each Sylow q -gp is cyclic, q prime and each finite abelian group is a direct sum of cyclic groups

① Sylow p -subgroup is cyclic

Claim: $1+p$ is an element of order p^{n-1} in $(\mathbb{Z}/p^n\mathbb{Z})^\times$
(Show that $(1+p)^{p^{n-1}} \equiv 1 \pmod{p^n}$ but $(1+p)^{p^{n-2}} \not\equiv 1 \pmod{p^n}$).

② Sylow q -subgroup is cyclic, $q \neq p$.

The map

$$\begin{aligned} \mathbb{Z}/p^n\mathbb{Z} &\rightarrow \mathbb{Z}/p\mathbb{Z} \\ a + (p^n) &\mapsto a + (p) \end{aligned}$$

is a ring homo which gives a surjective homo from $(\mathbb{Z}/p^n\mathbb{Z})^\times$ onto $(\mathbb{Z}/p\mathbb{Z})^\times$. The latter group is cyclic of order $p-1$. The kernel of this map is of order p^{n-1} , hence for all primes $q \neq p$, the Sylow q -subgroup of $(\mathbb{Z}/p^n\mathbb{Z})^\times$ maps isomorphically into the cyclic group $(\mathbb{Z}/p\mathbb{Z})^\times$. All Sylow subgroups of $(\mathbb{Z}/p^n\mathbb{Z})^\times$ are therefore cyclic, so $(\mathbb{Z}/p^n\mathbb{Z})^\times$ is cyclic.

$$\triangleright \text{Aut}(C_{2^e}) = C_{2^{e-2}} \times C_2 \quad (e \geq 2)$$

Note that C_2 and $\mathbb{Z}/4\mathbb{Z}$ have primitive roots (1 and 3 respectively), but not for $\mathbb{Z}/2^k\mathbb{Z}$.

① If $a = 2b+1$ for some b , $k \geq 3$, then $a^{2^{k-2}} \equiv 1 \pmod{2^k}$ when $k=3$. $a^{2^1} = (2b+1)^2 = 4b^2 + 4b + 1 = 1 + 4b(b+1) \pmod{2^3}$

Since $8 \mid 4b(b+1)$, $a^2 \equiv 1 \pmod{2^3}$.

Suppose it holds for $k \leq n$.

$$\text{Then } (1 + \ell \cdot 2^k)^2 = \ell^2 \cdot 2^{2k} + \ell \cdot 2^{k+1} + 1 \equiv 1 \pmod{2^{k+1}}.$$

② 5 is an element of order 2^{k-2} in $(\mathbb{Z}/2^k\mathbb{Z})^\times$

(show that $(1+2^2)^{2^{k-2}} \equiv 1 \pmod{2^k}$ and $(1+2^2)^{2^{k-3}} \equiv 1+2^{k-1} \not\equiv 1 \pmod{2^k}$)

③ Since $(\mathbb{Z}/2^k \mathbb{Z})^\times$ is finite abelian, it is a direct product of cyclic groups. It has an element of order 2^{k-2} , then $(\mathbb{Z}/2^k \mathbb{Z})^\times \simeq C_{2^{k-2}} \times C_2$.

▷ $\text{Aut}(\mathbb{Z}_n) = \text{Aut}(\mathbb{Z}_{p_1^{e_1}}) \times \dots \times \text{Aut}(\mathbb{Z}_{p_t^{e_t}})$ where $n = p_1^{e_1} p_2^{e_2} \dots p_t^{e_t}$

pf: Corollary of Chinese Remainder Theorem.

Thm (Chinese Remainder Theorem) $\checkmark$ (commutative with identity).

Let $A_1, A_2, \dots, A_k$ be ideals in R . The map

$$R \xrightarrow{\varphi} R/A_1 \times R/A_2 \times \dots \times R/A_k$$

defined by $r \mapsto (r+A_1, r+A_2, \dots, r+A_k)$

is a ring homo with kernel $A_1 \cap A_2 \cap \dots \cap A_k$.

If for each $i, j \in \{1, 2, \dots, k\}$ with $i \neq j$ the ideals A_i and

A_j are coprime ($A_i + A_j = R$) then this map is surjective

and $A_i \cap A_j \cap \dots \cap A_k = A_i \dots A_k$ so

$$R/(A_1 A_2 \dots A_k) = R/(A_1 \cap A_2 \cap \dots \cap A_k) \simeq \prod_{i=1}^k R/A_i$$

or simply

Thm (CRT)

Let $m_1, \dots, m_n$ be pairwise coprime integers. Then

$$\mathbb{Z}/(m_1 \dots m_n) \mathbb{Z} \simeq (\mathbb{Z}/m_1 \mathbb{Z}) \times \dots \times (\mathbb{Z}/m_n \mathbb{Z})$$

pf of general version:

① homo with $\ker \varphi = A_1 \cap \dots \cap A_n$ is obvious

② surjective if coprime

lemma R is a ring with identity. I_1, I_2, J are ideals of R .

If I_1 and J are coprime, then $I_1 I_2$ and J are coprime.

pf: Coprime

$$\exists a_1 \in I_1, b_1 \in J \text{ s.t. } a_1 + b_1 = 1$$

$$\exists a_2 \in I_2, b_2 \in J \text{ s.t. } a_2 + b_2 = 1$$

$$\text{Then } (a_1 + b_1)(a_2 + b_2) = 1 = \underbrace{a_1 a_2}_{\in I_1 I_2} + \underbrace{a_1 b_2 + b_1 a_2 + b_1 b_2}_{\in J}$$

By lemma, I_1 and $I_2 \dots I_n$ are coprime, so

$$\exists a_1 \in I_1, b \in I_2 \dots I_n \text{ s.t. } a_1 + b = 1$$

$$\text{let } x_1 = 1 - a_1 = b, \text{ then } x_1 + I_1 = 1 - a_1 + I_1 = 1 + I_1$$

$$x_1 + I_i = b + I_i = I_i, i \neq 1$$

$$\Rightarrow \varphi(x_1) = (1 + I_1, I_2, \dots, I_n)$$

Similarly find x_i and so $\forall x = (r_1 + I_1, \dots, r_n + I_n) \in \prod_{i=1}^n R/I_i$

$$\exists r = r_1 x_1 + \dots + r_n x_n \text{ s.t. } x = \varphi(r)$$

③ coprime then $A_1 \cap \dots \cap A_n = A_1 \dots A_n$

$\supseteq$ is obvious.

$\subseteq$ By induction, it suffices to prove $A_1 \cap A_2 \subseteq A_1 A_2$.

Indeed, if $a \in A_1 \cap A_2$, $x \in A_1, y \in A_2$ are such that $x + y = 1$. then $a = a(x + y) = \underbrace{ax}_{\in A_2 A_1} + \underbrace{ay}_{\in A_1 A_2} \in A_1 A_2$ since R

is commutative.

Since the isomorphism in the Chinese Remainder Theorem is an isomorphism of rings, in particular the groups of units on both sides must be isomorphic. Then

$$(\mathbb{Z}/mn\mathbb{Z})^\times \simeq (\mathbb{Z}/m\mathbb{Z})^\times \times (\mathbb{Z}/n\mathbb{Z})^\times$$

whenever $(m, n) = 1$.

Remark: 3 elements of order 2 in $\text{Aut}(\mathbb{C}^{2^e})$

$$6: g \rightarrow g^{-1}$$

$$7_0 = 7^{2^{e-3}}: g \rightarrow g^{2^{e-1}+1} \quad \text{i.e. use } 7_5: g \rightarrow g^5$$

$$67_0 = 67^{2^{e-3}}: g \rightarrow g^{2^{e-1}-1} \quad (= (g^{2^{e-1}+1})^{-1})$$

Dihedral group $G = D_{2n}$ 二面体群

$$G = \langle a, b \rangle, \quad a^2 = b^2 = 1$$

Ex

Let $a = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad b = \begin{pmatrix} 0 & 2 \\ \frac{1}{2} & 0 \end{pmatrix} \in GL(2, \mathbb{R})$.

$|a| = |b| = 2, \quad ab = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 2 \end{pmatrix}$ elements of order 2 are called involutions.

$G = \langle a, b \rangle$ is dihedral, infinite.

> Then $|G|$ finite $\Leftrightarrow |ab|$ finite

(elements of G are $a, ab, aba, abab, \dots$ and $b, ba, bab, baba, \dots$ as soon as we know the first term in these lists to give the identity element, we know G . It can't be elements like $ababa$, since $1 = ababa = bab = a$ so it must be $(ab)^m$ for some m then G is finite.

(note that $(ab)^m = 1 \Leftrightarrow (ba)^m = 1$).

$$> G = \{(ab)^i, (ba)^j, (ab)^i a, (ba)^j b\}$$

$$\Rightarrow G = \{(ab)^i, b(ab)^i \mid i\}$$

Since $(ba)^j = (ba)^j b^2 = b(ab)^j b = (ab)^j$

$$(ab)^i a = b(ba)^{i+1} = b(ab)^i$$

Then if G is finite, $|ab| = n$, so $|G| = 2n$ and $|ab| = \frac{|G|}{2}$

$$> G = \langle ab, b \rangle = \langle ab, a \rangle = \langle ab \rangle : \langle b \rangle = C_n : C_2$$

semidirect product

def let G be a gp, $A, B < G$

$$\textcircled{1} A < G \quad \textcircled{2} A \cap B = 1$$

$$\textcircled{3} G = AB$$

Then G is called a semidirect product of A by B denoted $A : B$ or $A \rtimes B$.

$$D \text{ Aut}(D_{2n}) \quad \langle Z = \begin{matrix} g \rightarrow g \\ b \rightarrow gb \end{matrix} \rangle \quad \left\{ \begin{matrix} 6_i \\ 6_i: g \rightarrow g^i, (i, n)=1 \\ b \rightarrow b \end{matrix} \right\}$$

(i) n is odd

$$\text{Aut}(D_{2n}) \cong C_n : \text{Aut}(C_n)$$

$(6 \in \text{Aut}(D_{2n}) :$

$$6: g \rightarrow g^i, (i, n)=1$$

$$b \rightarrow g^j b \quad 0 \leq j \leq n-1$$

$$|\text{Aut}(D_{2n})| = n \phi(n)$$

In particular, $\text{Aut}(D_{2p}) = C_p : C_{p-1}$

$$\text{Aut}(D_{2p^e}) = C_{p^e} : C_{p^e-1}^{(p-1)}$$

Hint: $G = D_{2n} = \langle g \rangle : \langle b \rangle = C_n : C_2$

If $6 \in \text{Aut}(G)$, then $g^6 = g^i$ since $\circ(g^6)$ is odd

▷ Conjugacy classes of elts of $D_{2n} = \langle g \rangle : \langle b \rangle$

① n is odd

$$= \{g^i, g^i b \mid 0 \leq i, j \leq n-1\}$$

(i) elts of odd order form

conjugacy classes $\{g^i, g^{-i}\}$

(ii) elts of even order are all conjugate.

pf:

Claim: $g^j b$ and $g^{j+1} b$ are conjugate.

Observe that

① D_{2n} has exactly n elts of order 2: $g^j b, 0 \leq j \leq n-1$

② $g^{-i} b g^i$ with $0 \leq i \leq n-1$ are of order 2 and distinct and are obviously conjugated. $\Rightarrow g^i b = g^{-j} b g^j$ for some j and $g^i b$ are conjugated.

③ n is even.

Week 2

Constructing gps and finding $\text{Aut}(G)$ for the new (G) .Construction

Let G be a group and let $H \leq \text{Aut}(G)$. Let $X = \{gh \mid g \in G, h \in H\}$ s.t. $(g_1 h_1)(g_2 h_2) = (g_1 g_2^{h_1^{-1}})(h_1 h_2)$. Then X is a gp. Denoted by $G \cdot H$ in which $G \triangleleft X$ and $X/G \cong H$.

Example cyclic group

Let $G = Z_7 = \langle g \rangle$. Then $\text{Aut}(G) = Z_6$.

① Let $\phi \in \text{Aut}(G)$ of order 2. ($g^6 = g^{-1}$)

Then $X = G \cdot \langle \phi \rangle = \langle g \rangle \cdot \langle \phi \rangle$ with $(g\phi)(g\phi) = gg^6 = 1$
 $\Rightarrow X = \langle g, \phi \mid g^6 = g^{-1}, |g|=7, |\phi|=2 \rangle = Z_7 \cdot Z_2 = D_{14}$.

② Let $\tau \in \text{Aut}(G)$ of order 3

Then $g^2 = g^{\tau}$

$\Rightarrow X = \langle g, \tau \rangle$ where $|g|=7, |\tau|=3, g^2 = g^{\tau} \xrightarrow{\text{nonabelian}} = Z_7 \cdot Z_3$

Example general cyclic group.

Let $G = \langle g \rangle = Z_p$. p prime. Then $\text{Aut}(G) = Z_{p-1}$

Let $\phi \in \text{Aut}(G)$ s.t. $|\phi| = l$. Then $l \mid (p-1)$ and $\langle \phi \rangle = Z_l$.

$X = G \cdot \langle \phi \rangle = Z_p \cdot Z_l = \langle g, \phi \rangle$ where $|g|=p, |\phi|=l, g^{\phi} = g^{\lambda}$

If $|\phi|=2$, then $X = D_{2p}$

$$(Z_7 : Z_6) = Z_7 \rtimes Z_6$$

$$(Z_7 : Z_3) \times Z_2$$

$$(Z_7 : Z_2) \times Z_3$$

$$Z_7 : Z_6$$

If $l = p-1$, then $X = Z_p : Z_{p-1}$

$$X = \text{AGL}(1, p) \Leftrightarrow Z(X) = 1$$

In general, $G = Z_n, X = Z_n \cdot H, H \leq U(n)$.

Example

$$G = Z_{35} = Z_5 \times Z_7. \quad H \leq \text{Aut}(G) = Z_4 \times Z_6 = \langle b_1 \rangle \times \langle b_2 \rangle$$

$$X = Z_{35} : H \leq (Z_5 : Z_4) \times (Z_7 : Z_6)$$

$$X_0 = Z_{35} : \langle b_1^2 b_2^3 \rangle = D_{70}$$

$$\text{Let } G = D_{2n} = Z_n : Z_2$$

$$\text{Then } \text{Aut}(G) = Z_n : U(n) = Z_n : \text{Aut}(Z_n).$$

Example finite field.

$$\text{Let } F = GF(p^f) \text{ and let } G = F^+ \cong Z_{p^f}, \quad H = F^\times \cong Z_{p^f-1}$$

Define a group $X = \{gh \mid g \in G, h \in H\}$ s.t. $(g_1, h_1)(g_2, h_2) = g_1 g_2^{h_1^{-1}} h_1 h_2$
 where $g_2^{h_1^{-1}} = g_2 \cdot h_1^{-1}$ in F
 $\downarrow$ multiplication.

Let $g, h \in F$. Then h is an auto of $G = F^+$ induced by multiplication by h . as $(a+b)^h = (a+b)h = ah + bh = a^h + b^h$

$$\Rightarrow X = G : H = Z_{p^f} : Z_{p^f-1} \quad \text{"Singer cycle"}$$

$$\text{Aut}(X) = X : \langle z \rangle = (Z_{p^f} : Z_{p^f-1}) : Z_f = (Z_{p^f} : \langle g \rangle) : \langle z \rangle$$

where $z : g \mapsto g^p$ "Frobenius" auto

$$\text{In particular, } \text{Aut}(Z_p : Z_{p-1}) \cong Z_p : Z_{p-1}$$

Example

Assume $p=3, f=2$

$$\text{Then } \text{Aut}(X) = (Z_3^2 : Z_8) : Z_2 \quad \text{where } z : g \mapsto g^3$$

$$\text{let } a = g^2 \text{ \& } b = g^2 \quad \text{let } \langle a, b \rangle = Q_8 = \{1, -1, i, -i, j, -j, k, -k\}$$

$$(b^2 = g^2 g^2 = g g^{z^2} z^2 = g^4 \quad |b|=4 \quad a^2 = b^2)$$

$$a^b = (g^2)^{-1} g^2 g^2 = g^6 = (g^2)^{-1} = a^{-1}$$

$$b^a = b^{-1}$$

$$\langle g \rangle : \langle z \rangle \rightarrow \langle g^2, gz \rangle$$

$\uparrow$
 $\uparrow$

$$\text{Aut}(X) = (Z_3^2 : Z_8) : Z_2 = Z_3^2 : (Z_8 : Z_2) \cong Z_3^2 : Q_8$$

$\downarrow$ since g

HW: all elts of $Z_3^2 : Q_8$ of order 3 are conjugate.

Group action

Def group action

let G be a group and Ω a set. We say that G acts on Ω if

① each elt g of G is a 1-1 map from Ω to Ω , the image of $w \in \Omega$ under g is denoted by w^g

② $w^1 = w \quad \forall w \in \Omega$.

③ $(w^{g_1})^{g_2} = w^{g_1 g_2}$

Example

① $GL(\Omega)$ acts on $\Omega = F^n$

② conjugation $\Omega = G$. $w^g = g^{-1}wg$ $C(g) = \{g^x \mid x \in G\}$ conjugacy class of g

③ right multiplication $\Omega = G$ $w^g = wg$.

Lemma

(N/C - lemma)

For $H < G$, we have $C_G(H) \trianglelefteq N_G(H)$

and $N_G(H) / C_G(H) \cong \text{Aut}(H)$.

In particular, if $H \trianglelefteq G$, $G / C_G(H) \cong \text{Aut}(H)$.

pf: $N_G(H)$ acts on H by conj. with kernel equal to $C_G(H)$.

Week 3

Group actions

G finite.

Thm (Cauchy)

If $p \mid |G|$, then $\exists g \in G$ s.t. $\text{ord}(g) = p$.

Let G act on a set Ω . Then G partitions Ω into orbits.

$$\Omega = \Omega_1 \dot{\cup} \Omega_2 \dot{\cup} \dots \dot{\cup} \Omega_m$$

Thm (orbit-stabilizer thm)

$$|G| = |\Omega_i| |G_{w_i}| \text{ where } w_i \in \Omega_i \text{ and } G_{w_i} = \{g \in G \mid w_i^g = w_i\}$$

Example conjugation action.

Assume $\Omega = G$ and G acts on Ω by conjugation. Then

$$G = \{1\} \cup C(x_1) \cup \dots \cup C(x_t)$$

$$\text{Then } |G| = |C(x_1)| + |C(x_2)| + \dots + |C(x_t)|. \text{ In particular, } |C(x_i)| \mid |G|$$

Thm If $|G| = p^e$ with p prime, then $Z(G) \neq 1$.

$$\text{pf: } G = \{1\} \cup C(x_1) \cup \dots \cup C(x_t)$$

$$\text{Then } |G| = p^e = 1 + |C(x_1)| + \dots + |C(x_t)|$$

$$= 1 + p^{e_1} + \dots + p^{e_t} \text{ where } e_i \geq 0.$$

It follows that $p^{e_i} = 1$ for some i . So $|C(x_i)| = 1 \Rightarrow 1 \neq x_i \in Z(G)$

Remark: If x and y are conjugate, then $|x| = |y|$.

Converse is not necessarily correct. i.e. In $C_m = \langle g \rangle$, g and g^{-1} are not conjugate.

Problem If G is a group satisfying that elts of the same order are conjugate, then $G = S_2$ or S_3 . (solved independently by ¹⁹⁸W. Feit - G. Thompson)

Sylow thms G finite, $|G| = p^e m$ with p prime and $\gcd(p, m) = 1$

Def Sylow p -subgroup

If G has a subgroup H s.t. $|H| = p^e$, then H is called a Sylow p -subgroup.

Remark existence of $H \Rightarrow$ Cauchy thm.

Thm (first Sylow thm)

Sylow p -subgroups exist $\forall p \mid |G|$.

pf: use conjugation equation.

$$G = \{1\} \cup C(x_1) \cup \dots \cup C(x_s)$$

$$|G| = 1 + |C(x_1)| + \dots + |C(x_s)| = Z(G) + |C(x_1)| + \dots + |C(x_s)|$$

① $p \nmid Z(G)$.

Then $\exists C(x_i)$ s.t. $(|C(x_i)|, p) = 1$ and $C(x_i) \neq 1$.

By orbit-stab thm, as $|G| = |C(x_i)| |C_G(x_i)|$, $p^e \parallel |C_G(x_i)|$ i.e. $C_G(x_i)$ is a subgroup of G of order divisible by p^e with $|C_G(x_i)| < |G|$.

By induction, $C_G(x_i)$ has a Sylow p -subgroup P . Then P is a Sylow p -subgroup of G .

② $p \mid Z(G)$.

$Z(G)$ is an abelian group. We first prove Cauchy's thm for abelian group.

Let G be abelian. $\forall p \mid |G|$. pick $1 \neq x \in G$. If $p \mid |x|$, then $x^{\frac{|x|}{p}}$ is of order p . If $p \nmid |x|$, then $|G/\langle x \rangle|$ is divisible by p . By induction, $\exists a \in G$ s.t. $a \langle x \rangle$ is of order p . then $p \mid |a|$. since $(a \langle x \rangle)^{\frac{|a|}{p}} = 1$, then $a^{\frac{|a|}{p}}$ is of order p .

Then $\exists x \in Z(G)$ is of order p . so $p^{e-1} \parallel (G/\langle x \rangle)$.

By induction, $G/\langle x \rangle$ has a subgroup of the form $H/\langle x \rangle$ of order p^{e-1} . Then H is a subgroup of order p^e . $\square$

Thm (Second Sylow thm)

Sylow p -subgroups are conjugate.

pf: Let $\Omega = \{P_1, P_2, \dots, P_n\}$ be the set of Sylow p -subgroups. Let $P = P_1$ and let $\Delta = \{Px \mid x \in G\}$. $|\Delta| = [G:P]$. Then G acts on Δ by right multiplication: $g: Px \mapsto Pxg$.

Then $|\Delta| = \frac{|G|}{|P|}$, coprime to p and the action of G on Δ is transitive.

Let $H < G$ s.t. $|H| = p^d \leq p^e$. Then H acting on Δ is intransitive. (Otherwise $m \mid p^d$, contradiction). Suppose H acting on Δ partitions Δ into orbits $\Delta_1, \dots, \Delta_r$. Then $|\Delta_i| \mid |H|$ and $|\Delta| = |\Delta_1| + \dots + |\Delta_r|$. So $|\Delta_i| = 1$ for some i since $p \nmid |\Delta|$ i.e. $Pxh = Px$ for some x , for each $h \in H$.

$$xHx^{-1} \subset P$$

$$H < x^{-1}Px$$

If $|H| = p^e$, $H = x^{-1}Px$ $\square$

Thm (Third Sylow thm)

Let $|G| = p^e m$ with $\gcd(p, m) = 1$. and let n_p denotes the number of Sylow p -subgroups of G . Then $n_p \mid m$ and $n_p \equiv 1 \pmod{p}$.

Group actions

Let G be finite. $1 \neq N \triangleleft G$. Then $\bar{G} = G/N$ is a group with $|\bar{G}| < |G|$. So $G = N \cdot \bar{G}$ (extension).

Lemma

If N is a minimal normal subgroup of G , then $N = S_1 \times \dots \times S_r$ where $S_1 \cong S_2 \cong \dots \cong S_r$ simple.

i.e. $N = S^r \cong S_1 \times \dots \times S_r$. Further $N = C_p^r$ or S is nonabelian.

pf:

If N is simple, then OK.

Assume that N is not simple. Let M be a minimal normal subgroup of N . By induction, $M = T_1 \times \dots \times T_m$ where $T_1 \cong T_2 \cong \dots \cong T_m$ simple.

For $g \in G$, the conj $M^g = g^{-1} M g$ is a minimal normal subgroup of N , either $M^g = M$ or $M \cap M^g = \{1\}$ by minimality.

The normal closure of M under G is

$$M^G = \langle M^g \mid g \in G \rangle = M \times M^{g_1} \times \dots \times M^{g_r}.$$

(Note that if $M \cap M^g = 1$, then $\langle M, M^g \rangle = M \times M^g$ since $M \triangleleft N, M^g \triangleleft N$).

Also, since $M = T_1 \times \dots \times T_m \Rightarrow M^g = T_1^g \times \dots \times T_m^g$.

Hence $M^g = T^{m \times r}$.

$M^G \triangleleft G$ and $M^G < N \Rightarrow N = M^G = T^{m \times r}$. $\square$

Now change notation: $M \triangleleft_{\min} G$. $M = \begin{cases} C_p^r \\ T_1 \times \dots \times T_r \text{ nonabelian} \end{cases}$

$$G/C_G(M) = N_G(M)/C_G(M) \cong \text{Aut}(M) \text{ where } M \triangleleft G.$$

(note that if $M \triangleleft G$ then $C_G(M) \triangleleft G$).

Q: For T_i nonabelian, want to find $\text{Aut}(T_1 \times \dots \times T_r)$

For $g \in \text{Aut}(M)$, we have $T_i^g \triangleleft M$ as $T_i \triangleleft M$.

Claim: $T_i^g = T_j$ for some j .

pf:

$$T_i^g \triangleleft T_1 \times \dots \times T_r = M$$

Suppose $T_i^g \not\subseteq T_2 \times \dots \times T_r \triangleq L$, then $T_i^g \cap L = 1$ since $T_i^g \cap L \triangleleft T_i^g$.

Then since $T_i^g \subset M = T_1 \times \dots \times T_r$, $\exists (t_1, y) \in T_i^g$ where $t_1 \neq 1$ and $y \neq 1, y \in L$.

$$(t_1, y)^{(t, 1)} = (t_1^t, y) \quad \forall t \in T_1.$$

Since T_1 is nonabelian and simple, $1 \neq Z(T_1) \triangleleft T_1$, hence

$$\exists t \in T_1 \text{ s.t. } t_1^t \neq t_1, \text{ and } (t_1, y)^{-1} (t_1^t, y) = (\underbrace{t_1^{-1} t_1^t}_{t_0}, 1) \in$$

$$\langle (t_0, 1)^{(t, 1)} \mid (t, 1) \in T_1 \times L \rangle \triangleleft T_i^g$$

$$\langle (t_0^t, 1) \mid t \in T_1 \rangle \triangleleft T_1 \Rightarrow \langle \rangle = T_1 \Rightarrow T_i^g = T_1.$$

On the other hand, $T_i^g \subset T_2 \times \dots \times T_r$, continue. $\square$

$\Rightarrow \text{Aut}(M)$ induces a permutation group on $\{T_1, T_2, \dots, T_r\}$ with kernel $\text{Aut}(T_1) \times \dots \times \text{Aut}(T_r) =: K$.

$$\text{s.t. } \text{Aut}(M) / K = S_r \Rightarrow \text{Aut}(M) = K \cdot S_r = \text{Aut}(T)^r \cdot S$$

Example

structure of G where $|G| = 72$

Suppose it is solvable, $\exists N \triangleleft_{\min} G$.

$$N \cong T^r \quad |T|^r \mid 72 \Rightarrow N \leq C_3^2 \text{ or } C_2^3$$

by conj. action

$$T \triangleleft \text{Aut}(T)$$

$$\text{Aut}(T) / T \cong \text{Aut}(T)$$

relatively small

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Date

Campus

Week 4

(Burnside)

If G is a finite group of which sylow subgps are all cyclic, then G is solvable.

Def transfer from G to H/H' ($H < G$)

G finite. $H < G$ $\Omega = [G : H] = \{Hx \mid x \in G\} = \{Hx_1, \dots, Hx_n$

$$\hat{g} : Hx_i \rightarrow Hx_j = Hx_i g$$

so $x_i g \in Hx_j$ and $x_i g = hx_j$

$\hat{g}$ induces a permutation on $\{1, 2, \dots, n\}$ denoted by $z(g)$.

i.e. $i^{z(g)} = j$

$$h_i(g) = x_i g x_j^{-1} = x_i g x_i^{-1} z(g) \in H$$

let

$$V_{G \rightarrow H} : G \rightarrow H/H'$$

$$g \mapsto \prod_{i=1}^n x_i g x_i^{-1} z(g) H' = \prod_{i=1}^n h_i(g) H'$$

called a transfer from G to H/H' .

Lemma

(1) $V_{G \rightarrow H}$ is a homomorphism from G to H/H'

(2) $V_{G \rightarrow H}$ does not depend on the choices of $x_1, \dots, x_n$.

(3) writing $\hat{g}$ in cycle form.

$$V_{G \rightarrow H}(g) = \prod_{i=1}^t (x_i g^{l_i} x_i^{-1}) H'$$

pf:

(1) For $g_1, g_2 \in G$

$$V_{G \rightarrow H}(g_1 g_2) = \prod_{i=1}^n h_i(g_1 g_2) H'$$

$$= \prod x_i g_1 g_2 x_i^{-1} z(g_1 g_2) H' \in H/H', \text{ abelian}$$

$$= \prod x_i g_1 x_i^{-1} z(g_1) x_i z(g_1) g_2 x_i^{-1} z(g_1) z(g_2) H'$$

$$= \prod x_i g_1 x_i^{-1} z(g_1) H' \prod x_i z(g_1) g_2 x_i^{-1} z(g_1) z(g_2) H' = V_{G \rightarrow H}(g_1) V_{G \rightarrow H}(g_2)$$

(2) Let $y_1, \dots, y_n \in G$ s.t. $H y_i = H x_i$, $1 \leq i \leq n$.

Then $y_i = t_i x_i$, for some $t_i \in H$, $1 \leq i \leq n$.

For $g \in G$, let $y_i g = k_i(g) y_i z(g)$. Then $k_i(g) = y_i g y_i^{-1} z(g)^{-1} \in H$.
The corresponding transfer.

$$\begin{aligned} \tilde{V}_{G \rightarrow H}(g) &= \prod_{i=1}^n k_i(g) H' = \prod_{i=1}^n y_i g y_i^{-1} z(g)^{-1} H' \\ &= \prod_{i=1}^n t_i x_i g (t_i z(g) x_i z(g))^{-1} H' \\ &= \prod_{i=1}^n t_i (x_i g x_i^{-1} z(g))^{-1} t_i^{-1} H' \\ &\stackrel{H/H' \text{ abelian}}{=} \prod_{i=1}^n x_i g x_i^{-1} z(g) H' \prod_{i=1}^n t_i H' t_i^{-1} \\ &= \prod_{i=1}^n x_i g x_i^{-1} z(g) H' \\ &= V_{G \rightarrow H}(g). \end{aligned}$$

(3) $\hat{g}$ is a permutation on $\Omega = \{H x_1, \dots, H x_n\}$.

$\langle \hat{g} \rangle$ partitions Ω into t orbits: $(H x_i g^{k_i-1} = H x_i)$

$(H x_1, H x_1 g, \dots, H x_1 g^{k_1-1}), (H x_2, H x_2 g, \dots, H x_2 g^{k_2-1}), \dots, (H x_t, \dots, H x_t g^{k_t-1})$

Take $x_1, x_1 g, \dots, x_1 g^{k_1-1}, x_2, x_2 g, \dots, x_2 g^{k_2-1}, \dots, x_t, x_t g, \dots, x_t g^{k_t-1}$ as reps.

The corresponding transfer is

$$\begin{aligned} V_{G \rightarrow H}(g) &= \prod x_i g (x_i g)^{-1} (x_i g) g (x_i g)^{-1} \dots x_i g^{k_i-1} g x_i^{-1} H \\ &= \prod (x_i g^{k_i} x_i^{-1}) H' \quad \square. \end{aligned}$$

Def p -nilpotent

A group G is called p -nilpotent if $G = NP$ where $N \triangleleft G$ and P is a Sylow p -subgroup and $N \cap P = 1$.

($\Leftrightarrow$ If G has a normal subgp N s.t. $|G/N| = |G|_p = |P|$ where p is a prime divisor of $|G|$ and P is any Sylow p -subgroup of G).

Thm (Burnside)

Let G be a finite group and P a Sylow p -subgp. If $N_G(P) = C_G(P)$, then G is p -nilpotent.

pf: (use transfer to construct $\ker V_{G \rightarrow P} = N$.)

Since $N_G(P) = C_G(P)$, P is abelian ($P \leq N_G(P)$ hence $P \leq C_G(P)$), so $P' = 1$.

Let $g \in P \setminus \{1\}$. Then

$$V_{G \rightarrow P}(g) = \prod_{i=1}^t x_i g^{l_i} x_i^{-1}$$

As $g^{l_i}, x_i g^{l_i} x_i^{-1} \in P$ and P is abelian, $C_G(g^{l_i}) \geq P$, $C_G(x_i g^{l_i} x_i^{-1}) = x_i C_G(g^{l_i}) x_i^{-1} \geq P$. Hence $C_G(g^{l_i}) \geq P$ and $C_G(g^{l_i}) \geq x_i^{-1} P x_i$.

By Sylow thm, there exists $u \in C_G(g^{l_i})$ s.t.

$$u^{-1} (x_i^{-1} P x_i) u = P$$

||

$$(x_i u)^{-1} P (x_i u) = P$$

Thus $x_i u \in N_G(P) = C_G(P) \Rightarrow x_i \in C_G(P) \leq C_G(g^{l_i})$

Therefore $V_{G \rightarrow P}(g) = \prod_{i=1}^t g^{l_i} = g^{l_1 + \dots + l_t} = g^{|G:P|} \neq 1$

(since $g \in P \setminus \{1\}$)

which yields $V_{G \rightarrow P}(P) = P$ (since it is injective by above discuss

and $|G| < \infty \Rightarrow V_{G \rightarrow P}(G) = P$.

let $\ker V_{G \rightarrow P} \triangleq K$. Then by iso thm $G/K \cong P$ so $K \triangleleft G$

Thm (Burnside)

Let G be a finite group of which Sylow subgroups are all cyclic. Let p be the smallest prime divisor of G and let P be a Sylow p -subgroup of G .

Then (1) G is p -nilpotent

(2) G is solvable.

pf: Now P is cyclic and $N_G(P)/C_G(P) \leq \text{Aut}(P)$
 where $|\text{Aut}(P)| = p^{e-1}(p-1)$ whose prime divisor $\leq p$.
 Then we can only have $N_G(P)/C_G(P) = 1$ since $p^e \parallel |G(P)|$
 as $P \leq C_G(P)$, all prime divisors of LHS $> p$.
 Then $N_G(P) = C_G(P)$ and so G has a normal subgroup K
 with $|K| = |G|/|P|$.

Let p_1 be the smallest prime divisor of $|K|$. Then K is
 p_1 -nilpotent and has a normal subgroup K_1 with $|K_1| = |K|/|P_1|$.

Repeating this process, we have that

$$G \triangleright K \triangleright K_1 \triangleright \dots \triangleright Q$$

Sylow q -subgp. where q is the
 largest prime divisor of $|G|$.
 $\square$

Remark $Q \triangleleft G$.

pf: Suppose $Q \triangleleft K_n \triangleleft K_{n-1}$.

Then $\forall g \in K_{n-1}$, $g^{-1}Qg \subseteq g^{-1}K_n g = K_n$. Since $Q \triangleleft K_n$,
 K_n has only one Sylow q -subgroup, then $g^{-1}Qg = Q$, hence $Q \triangleleft K_{n-1}$.
 Repeat this process, $Q \triangleleft G$.

Corollary

If a Sylow 2-subgroup of G is cyclic, then G is 2-nilpotent
 hence solvable. ($\Rightarrow$ any Sylow 2-subgroup are cyclic).

pf: need to use Feit-Thompson: $|G|$ odd $\Rightarrow G$ solvable
 Then N & G/N both solvable $\Rightarrow G$ solvable.

Prop

Let G be a finite nonabelian simple group. Let p be a prime divisor of $|G|$.

(1) If p is the smallest prime divisor, then G is divisible by p^3 or 12.

(2) If $|G| = pm$ with $\gcd(p, m) = 1$. Then for a Sylow p -subgrp P ,

(i) $C_G(P) \neq N_G(P) \leq G$

(ii) $|N_G(P)/C_G(P)| \mid (p-1)$.

pf:

(1) Suppose $|G|$ is not divisible by p^3 . Then $p^2 \parallel |G|$ otherwise a Sylow p -subgrp $P = C_p$ so that $N_G(P) = C_G(P)$ and G is p -nilpotent, contradiction. $\rightarrow$ since center nontrivial

Then $|P| = p^2$ and so $P \cong C_{p^2}$ or $C_p \times C_p$. The former would imply G is p -nilpotent, not possible. Hence $P \cong C_p \times C_p$ and $N_G(P) \neq C_G(P)$. So $N_G(P)/C_G(P) \cong \text{Aut}(P) = \text{Aut}(C_p \times C_p) \cong GL(2, p)$. (multiply by scalars = addition).

Now $|GL(2, p)| = p(p^2-1)(p-1) = p(p+1)(p-1)^2$

let r be a prime divisor of $|N_G(P)/C_G(P)|$. Then $r \mid p(p+1)$.

And $r > p$ ($P \cong C_p \times C_p$ so $P \leq C_G(P)$) so $p=2$ and $p+1=3$.

Hence $12 \mid |G|$.

(2) easy.

Example If $|G| = 5 \cdot 7 \cdot 13$, then G is cyclic

Let G_p denotes a Sylow p -subgroup.

• $G_{13} \triangleleft G$ ($N_G(G_{13})/C_G(G_{13}) \leq \text{Aut}(G_{13}) = C_{12}$)
 $\rightarrow N_G(G_{13}) \leq C_G(G_{13})$

• G is 5-nilpotent.

$\rightarrow G$ is 13-nilpotent. $G = G_{13} \times G_{5,7}$

Week 5

solvable

Let G' be the commutator subgroup of G .

Let $G'' = G^{(2)} = (G')'$ and $G^{(n)} = (G^{(n-1)})'$

$$G \triangleright G' \triangleright G'' \triangleright \dots \triangleright G^{(n)} \triangleright \dots$$

example G' of G

① G simple & G not abelian (G/G' abelian & $G \neq 1$) $\Rightarrow G' = G$

② $G = S_n$. $\begin{cases} G' = A_n & \text{if } n \geq 5 \\ G' = A_4 & \text{if } n = 4 \end{cases}$ (A_n generated by all 3-cycles
(23) = $(12)(13)(12)^{-1}(13)^{-1}$)

If G is finite,

$$G \triangleright G' \triangleright G^{(2)} \triangleright \dots \triangleright G^{(n)} \triangleright G^{(n+1)}$$

either $G^{(n)} = G^{(n+1)} = G^{(n+2)} = \dots = G^{(\infty)}$ or $G^{(n)} = 1$

notation.

Def solvable & perfect

(1) G is solvable if $G^{(n+1)} = 1$ for some n .

(2) G is called a perfect group if $G = G'$

example perfect groups

① all simple groups (nonabelian)

② A_n with $n \geq 5$. $(123)(345)(123)^{-1}(345)^{-1} = (143)$

③ $SL_d(q)$ when nonsolvable i.e. $d \geq 2, (d, q) \neq (2, 2), (2, 3)$
 $\downarrow Z(SL_d(q)) = Z(d, q-1)$.

prop properties of being solvable

(1) G is solvable, then so is each subgroup and each

factor group of G .

(2) If $N \triangleleft G$ s.t. N and G/N are solvable, then G is solvable (G' is in N or G/N)

(3) p -groups are solvable. (use (2) and induction and $Z(G) \triangleleft G$ is nontrivial)

Lemma If P is a nonsylow p -group of G , then $P \not\subseteq N_G(P)$

Def double coset (双陪集)

$$P < G.$$

$PxP = \{h_1xh_2 \mid h_1, h_2 \in P\}$. a double coset of P in G

Lemma # of right cosets of P contained in PxP is equal to $|P| / |P^x \cap P|$

pf: $PxP = \bigcup_{h \in P} Pxh$

$$Pxh_1 = Pxh_2 \iff h_1h_2^{-1} \in P^x \implies h_1h_2^{-1} \in P^x \cap P$$

pf of lemma:

Decompose $G = \bigcup_{x \in G} PxP$ (Note that for $x, y \in G$, $PxP = PyP$ or $PxP \cap PyP = \emptyset$).

• $PIP = P$ has a unique coset of P .

$$|G| = |PIP| + |Px_1P| + \dots + |Px_tP|$$

consider # of cosets of P on both sides.

LHS: divisible by P

RHS: $|Px_iP| = |P| / |P^{x_i} \cap P|$ is either divisible by p or is 1.

Then for some x_j , $|P^{x_j} \cap P| = 1$ $x_j \notin P$, so $P^{x_j} \cap P = \{1\}$ and $P^{x_j} = P$ i.e. $x_j \in N_G(P)$, so $P \subseteq N_G(P)$.

Thm subgps & normal subgps of p -groups.

Let G be a p -group

(1) If $H < G$, then $H < N_G(H)$.

(2) If $H \leq_{\max} G$, then $H \triangleleft G$ and $|G|/|H| = p$ (consider G/H) nontrivial subgp would give $H \neq K$

(3) G is solvable (find $H \leq_{\max} G$, use induction).

(4) $Z(G) > 1$

(5) If N is a normal subgroup of G and $|N| = p$, then $N \leq Z(G)$.

pf of (5):

By NC lemma,

$$G/C_G(N) = N_G(N)/C_G(N) \leq \text{Aut}(N) = \mathbb{Z}_{p-1}.$$

Then $G = C_G(N)$ so $N \leq Z(G)$.

Corollary A group of order p^2 with p prime is abelian.

pf: $Z(G)$ is nontrivial.

Q: what about $|G| = p^3$? (D_8, Q_8 not abelian).

Thm structure of G when $|G| = p^3$

$|G| = p^3$. Then either G is abelian (3 types) or

(1) for $p = 2$, $G = D_8$ or Q_8

(2) for $p \geq 3$, $G = \mathbb{Z}_{p^2} : \mathbb{Z}_p$ or $G = \langle a, b, c \mid a^p = b^p = c^p = 1, [a, b] = c \rangle$ ("exponent" = p) $p \nmid H_2$

$\downarrow$ max. $\{|g| \mid g \in G\}$.

pf: Assume G is nonabelian.

① suppose G has an element a of order p^2

By above thm, $\langle a \rangle \triangleleft G$. Then there is $b \in G \setminus \langle a \rangle$ and

abelian

D8.

$$j^{-1}ij = i, \quad \chi^2 i^3$$

$b^p \in \langle a \rangle$. ($\langle b \rangle$ is of order p in $G/\langle a \rangle$).

If $b^p = 1$, then $G = \langle a \rangle : \langle b \rangle$ as in (1) for $p=2$, (2) for $p>2$.

Suppose $|b| = p^2$. Then $b^p = a^\lambda$, $a^b = a^\lambda$ for some integer λ .

If $p=2$, then $a^b = a^\lambda = a^{-1}$ and $b^a = b^\lambda = b^{-1}$ ($a^b = a, a^2 \neq 1$).

$G = \langle a, b \mid a^4 = b^4, a^b = a^{-1}, b^a = b^{-1} \rangle \rightarrow$ unique elt of order:

$$\parallel \quad \text{D8.} \quad a^2 = b^2.$$

Suppose p is odd. Then $a^b = a^{1+kp}$ for some k with

$$1 \leq k \leq p-1 \quad (a^{b^p} = a = a^{\lambda^p} \Rightarrow \lambda^p \equiv 1 \pmod{p^2} \Rightarrow \lambda^p \equiv 1 \pmod{p})$$

$$\Rightarrow \lambda^p \equiv \lambda \equiv 1 \pmod{p} \Rightarrow \lambda = 1+kp.$$

Claim: $(ba^{-\lambda})^p = 1$.

$$ba^{-\lambda} \cdot ba^{-\lambda} = b^2 b^{-1} a^{-\lambda} b a^{-\lambda} = b^2 a^{-2\lambda - \lambda kp}.$$

$$(ba^{-\lambda})^p = b^p a^{-p\lambda - \lambda kp^2} = 1.$$

let $b' = ba^{-\lambda}$ as in (2).

② Suppose G does not have elts of order p^2 .

Then each non identity elt of G is of order p ($\Rightarrow p$ must be odd since if $\exists$ elt of G is of order 2 then G is abelian). Consider G' .

Since $Z(G)$ is nontrivial, pick $z \in Z(G)$, then $Z_p = \langle z \rangle \triangleleft G$.

$|G/\langle z \rangle| = p^2 \Rightarrow G/\langle z \rangle$ is abelian by above discussion.

Hence $G' \leq \langle z \rangle \Rightarrow G' = Z_p = \langle [a, b] \rangle$.

let $c = [a, b]$. Then $a^p = b^p = c^p = 1$, $c = [a, b]$ as in (2).

$\langle a, b \rangle < G$, $|\langle a, b \rangle| \neq p$ and p^2 since a, b does not commute so $\langle a, b \rangle = G$. $\square$

★ consider $\text{Aut}(G)$ of $|G| = p^3$

$$\text{Aut}(D_8) \simeq D_8$$

$$\text{Aut}(Q_8) \simeq S_4$$

$$\text{Aut}(\mathbb{Z}_{p^2} : \mathbb{Z}_p) \simeq P : \mathbb{Z}_{p-1}$$

$\downarrow$ p-group

$$\text{Aut}(P_+^{H^2}) \simeq GL(2, p).$$

e.g. $G = P_-^{H^2}$ or $P_+^{H^2}$

$$G' = Z(G) = \mathbb{Z}_p.$$

Let $z \in \text{Aut}(G)$. Then $(G')^z = G'$ ($(Z(G))^z = Z(G)$) and

z induces an automorphism $\bar{z}$ of $G/G' \simeq \mathbb{Z}_p \times \mathbb{Z}_p$:

① $G = \langle a \rangle : \langle b \rangle = \mathbb{Z}_{p^2} : \mathbb{Z}_p$ then $G' = \langle a^p \rangle$

This is because $a^b = a^{\bar{1}} \Rightarrow a^{b^p} = a^{\bar{1}^p} = a^{\bar{1}} \Rightarrow \bar{1} = 1 + lp \Rightarrow (a^p)^b = a^{p\bar{1}} = a^p$

Then $G/G' = \langle \bar{a}, \bar{b} \rangle = \mathbb{Z}_p \times \mathbb{Z}_p$ (abelian) s.t. $\bar{a}^z \in \langle \bar{a} \rangle$

$$\Leftrightarrow (|z|, p) = 1 \text{ then } |z| \mid (p-1)$$

② $G = P_+^{H^2}$ then $\bar{z} \in \text{Aut}(\langle \bar{a} \rangle \times \langle \bar{b} \rangle)$

z fixes $G' = \langle [x, y] \mid x, y \in G \rangle$ — 双线性型

$$z \in Sp(2, p) \simeq SL(2, p) \text{ extension scalar multiplication}$$

not required

Week 6. Free groups and minimal normal subgroup.

Free groups

Def free group.

→ (inverse set).

Let $\{x_1, \dots, x_n\}$ be a set and $\{x_1', \dots, x_n'\}$ s.t. $\{x_1, \dots, x_n\} \cap \{x_1', \dots, x_n'\} = \emptyset$. Let $S = \{x_1, \dots, x_n, x_1', \dots, x_n'\}$.

Let X be the set of all words of finite sequence compounds of the elements of S . For $u, v \in X$,

$$u = x_{i_1} \dots x_{i_r},$$

$$v = y_{j_1} \dots y_{j_t},$$

another way of definition.

define ① $u \cdot v = x_{i_1} \dots x_{i_r} y_{j_1} \dots y_{j_t}$

→ "reduced words":

② If $u = u_1 x_i$ and $v = x_i' v_1$, then $uv = u_1 x_i x_i' v_1 = u_1 v_1$.
 x_i and x_i' are no adjacent

This identification is an equivalent relation. which gives rise to an equivalent partition of X . Let $[uv]$ denote the class containing uv . Then $[u]$ denotes the class containing u (say $u, u x_i x_i', u x_i x_i', \dots$) and $[u][v] = [uv]$.

Then $\bar{X} = \{[u] \mid u \in X\}$ forms a group, called a free group generated by $x_1, \dots, x_n$.

i.e. $\bar{X} = \langle x_1, \dots, x_n \rangle$, denoted by F_n where n is to be called the rank of F_n .
all the reduced words form a group. product is also defined to be a reduced word.

Example $X = \langle x_1, x_2 \rangle$

If $g \in X$. then $g = x_1^{m_1} x_2^{m_2} x_1^{m_3} \dots x_2^{m_t}$
 or $x_2^{m_1} x_1^{m_2} \dots x_1^{m_t} \dots$ where $|m_i| \geq 1$

① $X = \langle x_1, x_2 \mid x_1^2 = 1, x_2^2 = 1 \rangle$ — infinite dihedral group

② $X = \langle x_1, x_2 \mid x_1^2 = x_2^2 = 1, [x_1, x_2] = m \rangle$ — D_{2m}

③ $X = \langle x, y \mid |x| = m, |y| = n, [x, y] = l \rangle$ — triangle group

$T_{m,n,l}$

球面上的三角形
内角和 $> 180^\circ$

纬线
经线

★ triangle group $T_{m,n,l}$

consider $(\frac{1}{m} + \frac{1}{n} + \frac{1}{l})\pi$:

- $> \pi$ spherical
- $= \pi$ Euclidean
- $< \pi$ hyperbolic

WLOG. assume $m \leq n \leq l$. then if $G \neq S_3$, $l > 1$

Case 1 $\frac{1}{m} + \frac{1}{n} + \frac{1}{l} > 1$

$\begin{cases} m=1 \\ m=2 \end{cases} \quad G = \langle y \rangle \text{ cyclic} \Rightarrow G = \mathbb{Z}_n.$

$n=2 \Rightarrow G = D_{2l}$

$n=3 \Rightarrow l=3 \text{ or } 4 \text{ or } 5.$

Further analysis shows $G = A_4, S_4$ or A_5

对应球面上
几何结构的对称。

正四面体
对称

立方体
(与正四面体对称)

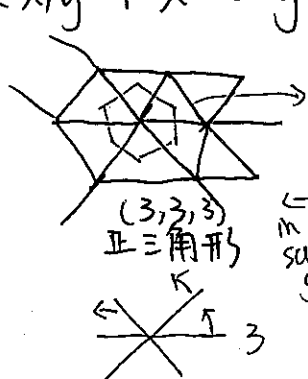
正十二面体

(对偶: 每个面的中心连起来)。

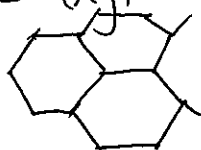
Case 2 $\frac{1}{m} + \frac{1}{n} + \frac{1}{l} = 1$

$(m, n, l) = (3, 3, 3), (2, 3, 6) \text{ or } (2, 4, 4)$

$G = \langle x, y \mid x^m = y^n = (xy)^l = 1 \rangle$ infinity, solvable.

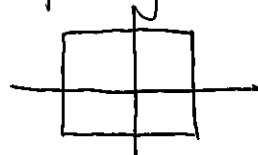
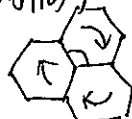


in fact
same
group.



(2,3,6)
正六边形

(正三角形)的对偶



正方形 (2,4,4)

(只有这三种正多边形能把平面铺满)。

Case 3 $\frac{1}{m} + \frac{1}{n} + \frac{1}{l} < 1$

Campus

$G = \langle x, y \mid x^m = y^n = (xy)^l \rangle$ infinity, nonsolvable

Example

$$S_n = \langle (12), (12 \dots n) \rangle$$

$$\Rightarrow S_n = \langle x, y \mid x^2 = y^n = 1, (xy)^{n-1} = 1, \dots \rangle$$

must be more.

otherwise by above discussion

$$1 > \frac{1}{2} + \frac{1}{n} + \frac{1}{n-1} \quad n \text{ large}$$

$\Rightarrow$ infinite. contradiction.

Thm Every f.g. group is a homomorphic image of a free group.

Let $G = \langle g_1, \dots, g_n \rangle$. Then F_n is homomorphic to G .

Pf: $\varphi: F_n \rightarrow G$

$$[x_i] \mapsto g_i$$

$$[x_i]^{-1} \mapsto g_i^{-1}$$

$$[\emptyset] \mapsto 1$$

let $N = \ker \varphi = \{x \mid \varphi(x) = 1\}$

Then $F_n / N \cong G$.

We then call $\{g_1, \dots, g_n\}$ generating elements (or generators) & $\{x \mid \varphi(x) = 1 \mid x \in N\}$ generating relations

e.g. $D_{2m} = \langle x, y \mid x^2 = y^2 = (xy)^m = 1 \rangle$

$$= \langle a, y \mid a^m = y^2 = 1, a^y = a^{-1} \rangle$$

Minimal Normal Subgroups

$$\Omega = \{1, 2, \dots, n\}$$

transitive in each orbit

If $H < \text{Sym}(\Omega) \cong S_n$ is intransitive, then $H \leq S_k \times S_{n-k}$

where $k \leq \frac{n}{2}$

let $G \leq \text{Sym}(\Omega) = S_n$.

Def quasi-primitive group.

A group G is called a quasi-primitive group if all its nontrivial normal subgroups are transitive.

Counterexample

$$G = D_{2n} < S_n$$

$$= \langle \underbrace{(1\ n)(2\ n-1) \cdots}_y, \underbrace{(1\ 2 \cdots n)}_x \rangle$$

when n is even, $\{b, a^2b, \dots, a^{n-2}b\} \triangleleft G$, but it has $\frac{n}{2}$ elements so it can not be transitive.

prop let G be a quasi-primitive group. let M be a minimal normal subgroup of G . then

(1) M is transitive. (if G any group, this is not always true.
e.g. $G = \{(12)(34), (13)(24), (14)(23), 1\} < S_4$

$$A = \{1, (12)(34)\} \triangleleft G \text{ not transitive}$$

(2) $M = T^l \cong T_1 \times \cdots \times T_l$ where $l \geq 1$, $T_1 \cong \cdots \cong T_l \cong T$ are simple.

Lemma If $M \triangleleft G$ is transitive, then $G = MG_w$ for any $w \in \Omega$.

pf: Given $g \in G$, let $w' = w^g$. Since M is transitive on Ω ,
 $\exists m \in M$ s.t. $w^m = w^g = w'$. Thus $mg = g \in Gw$
i.e. $g = my \in MG_w$ $\square$.

Lemma Let G be a quasi-primitive group. Then G has at most two minimal normal subgroups. If M, N are two minimal normal subgroups, then $M \trianglelefteq N$, nonabelian, regular.

pf: G has at most two minimal normal subgroups — to be proved

If M, N are two minimal normal subgroups, $M \cap N = \{1\}$.

So $M \times N = MN \triangleleft G$ and M and N commutes.

① regular

since N transitive.

For any $w, w' \in \Omega$, $\exists x \in N$, s.t. $w^x = w'$. Then $Mw' = Mw^x = x^{-1}Mwx = Mw$ since M and N commutes. So Mw fixes all points $w' \in \Omega$. Then $Mw = 1$ since M faithful. Similarly $Nw = \{1\}$.

② nonabelian

If M or N abelian, WLOG. assume M abelian. Then since M, N commute, $\langle M, x \rangle = H$ for any $x \in N$ is strictly bigger than M and abelian. It is transitive, then it is regular (since $Hw = Hw^x = x^{-1}Hwx = Hw'$). But for regular groups A , $\forall g, h \in \Omega$, $\exists! x \in A$ s.t. $g^x = h$.

Then $|A| = |\Omega|$. Since M and N regular, $|M| = |N| = |\Omega|$.

Hence $|H| > |\Omega|$ contradiction. Thus M and N both nonabelian.

③ $M \trianglelefteq N$

Let $X = M \times N \triangleleft G$.

Then $X = MXw = M : Xw$ and $Xw = \{(x, x') \mid x \in M, x' \in N\}$ where x' is uniquely determined by x .

Define $\varphi: M \rightarrow N$
 $x \mapsto y$ if $(x, y) \in Xw$.

prove ① φ is a homomorphism.

② $\ker \varphi = \{1\}$.

Week 7

Lemma

Let $G \leq \text{Sym}(\Omega)$ be transitive. Then $C_{\text{Sym}(\Omega)}(G)$ is semiregular.

pf: Let $C = C_{\text{Sym}(\Omega)}(G)$.

For $w, w' \in \Omega$, $\exists g \in G$ s.t. $w^g = w'$. Then $Cw' = Cw^g = Cw^g = g^{-1}Cwg = Cw$.

So $Cw = 1$ and C is semiregular.

Corollary

A transitive abelian permutation group G is regular.

pf: $G \leq C_{\text{Sym}(\Omega)}(G)$.

By lemma, since $M \leq C_{\text{Sym}(\Omega)}(G)$, $\text{Sym}(\Omega)(N)$ semiregular, $\uparrow M = C_{\text{Sym}(\Omega)}(N)$.

Thm (from last time)

Let G be a quasi-primitive group. Then G has at most two minimal normal subgs. If G has two minimal normal subgroups M, N , then $M \cong N$, M, N regular, nonabelian.

pf:

① M, N regular (use lemma).

$$X = MN = M \times N$$

$$\Rightarrow M \leq \text{Sym}(\Omega)(N)$$

$\Rightarrow M$ is regular, so is N .

② $M \cong N$.

for $w \in \Omega$, $Xw = \{ (x, x') \mid x \in M, x' \in N \} \leq M \times N$.

let $\varphi: M \rightarrow N$

$$x \mapsto x'$$

— surjective & injective

— isomorphism.

□.

Lemma

pf: Since N is nonabelian. $N \not\subseteq C_G(N)$. Since $C_G(N) \triangleleft G$ it would contain a minimal normal subgroup. if $C_G(N) \neq \{1\}$, contradiction. Thus $C_G(N) = \{1\}$. $\square$

Thus

ii) N is regular $\Leftrightarrow N_w = 1$ TW (twisted wreath product).

(ii) $N_W \cong T$ $\Rightarrow$ SD (simple diagonal)

(iii) $N_w \approx J^m$ $2 \leq m \leq \frac{k}{2}$ CD (compound diagonal)

(iv) $1 \neq N_W \not\subseteq T^m$ for $m \geq 1$. PA (product action).

Thm (O'Nan - Scott - Praeger)

Let G be a quasi-primitive group. Then one of the following holds:

(1) G has exactly two minimal normal subgroups $M \cong N$ simple HS (holomorph simple).

(2) G has two minimal normal subgroups $M \cong N \cong T^k$ with $k \geq 2$. H.C. type. (holomorph compound). holomorph affi

(3) G has a ^{unique} minimal normal abelian subgroup type HA ^{abeli}

4) G has a unique minimal normal subgroup N , simple and nonabelian. type AS (almost simple).

(57). G has a minimal normal nonabelian regular subgroup type IV

(6) ③ (ii) (above) (7) ③ (ii) (8) ③ (iv).

Case 11)

$G \leq \text{Sym}(\Omega)$. quasi-primitive.

let $M, N \triangleleft G$. Then $M \cong N$. nonabelian. regular. so $M \times N \cong G$
 $|\Omega| = |M| = |N|$.

fix $w \in \Omega$. for any $w' \in \Omega$. $\exists! x \in M$ s.t. $w^x = w'$
 $\Rightarrow$ There is a 1-1 correspondence between Ω and M by

$$\begin{aligned} w &\mapsto 1 \\ w' &\mapsto x \end{aligned}$$

Relabel :

$$\begin{aligned} \Omega &= \{w_1, w_2, \dots, w_n\} \quad \text{where } \Omega \rightarrow M \quad \text{with } w_i \mapsto x_i \quad \text{with } w_i = w_1 x_i \\ M &= \{1=x_1, x_2, \dots, x_n\} \end{aligned}$$

Lemma relationship between $\hat{g}, \check{g}, \tilde{g}$ (right multiplication / l.m. / conjugation)

let G be a finite group. fix $g \in G$. define permutations on G :

$$\begin{aligned} \hat{g} : x &\mapsto xg \\ \check{g} : x &\mapsto g^{-1}x \\ \tilde{g} : x &\mapsto g^{-1}xg \end{aligned} \quad \Rightarrow \quad \begin{aligned} \hat{G} &= \{\hat{g} \mid g \in G\} \\ \check{G} &= \{\check{g} \mid g \in G\} \\ \tilde{G} &= \{\tilde{g} \mid g \in G\} \end{aligned}$$

- ① $\hat{G}, \check{G} \leq \text{Sym}(G)$, regular
- ② $\langle \hat{G}, \check{G} \rangle = \hat{G}\check{G} = \hat{G} \circ \check{G}$ (central product) $\hat{G} \cap \check{G} = Z(\hat{G}) = Z(\check{G}) = Z(G) = Z$
- ③ let $X = \hat{G}\check{G}$, $X_1 = \hat{G}$ and $X = \hat{G} : \tilde{G} = \check{G} : \tilde{G}$

Def central product $(M \circ N \text{ (or } M * N))$:

$$C_1 \leq Z(M), \quad C_2 \leq Z(N).$$

$$C_1 \not\cong C_2, \quad C_1 \times C_2 \triangleleft M \times N$$

$$C = \{(C, C^x) \mid C \in C_1\} \cong C_1$$

$$M \circ N = M \times N / C \rightarrow M \circ_c N \quad (\text{usually when } C \text{ is maximal i.e. } Z(M) \cong Z(N) \cong C, \text{ called standard c.p. otherwise ad})$$

example

① $Q_8 \times Q_8$

$Z = \{1, -1\} = Z_2$

Let $Z = \langle C_1 \rangle = \langle C_2 \rangle$.

Then $Q_8 \times Q_8 / \langle (C_1, C_2) \rangle = Q_8 \circ Q_8$ with $|Q_8 \circ Q_8| = 32$.

② $Q_8 \circ Z_4$

$Q_8 \circ Z_4 = Q_8 \times Z_4 / \langle (C_1, C_2) \rangle$

③ $M = Z_8 : Z_2$
 $\langle a \rangle \quad \langle z \rangle$

$a^2 = a^5$

$(a^2)^2 = a^{10} = a^2$

$C_1 \in Z(M) = \langle a^2 \rangle = Z_4$

$N = \langle b \rangle : \langle y \rangle$
 $8 \quad 2$

$b^4 = b^5$

$C_2 \in Z(N) = Z_4$

— If $|C_1| = |C_2| = 4$

$M \times N / \langle (C_1, C_2) \rangle = M \circ N \rightarrow$ standard central product

— If $|C_1| = |C_2| = 2$

$M \times N / \langle \frac{(C_1, C_2)}{C} \rangle = M \circ_c N$

pf of lemma :

① $\hat{G}$ is a group as $\hat{g}, \hat{h} \in \hat{G} \Rightarrow \hat{g}\hat{h} = \widehat{gh} \in \hat{G}$

Similarly $\check{G}$ is a group. $(x\check{g}\check{h}) = (x\check{g})\check{h} = (g^{-1}x)\check{h} = h^{-1}g^{-1}x = x\check{g}\check{h}$

Moreover, for any $x, y \in \Omega (= G)$, $\hat{g} = x^{-1}y$ maps x to y so $\hat{G}$ transitive.

$\Rightarrow \hat{G}$ regular since $|\hat{G}| = |G|$.

② Let $\hat{g} \in \hat{G}$ and $\check{h} \in \check{G}$

Claim: $\hat{g}\check{h} = \check{h}\hat{g} \in \text{Sym}(\Omega)$.

For $x \in G$, we have $x \hat{g} \check{h} = (x \hat{g}) \check{h} = (xg) \check{h} = h x g$
 $x \check{h} \hat{g} = (x \check{h}) \hat{g} = (h^{-1} x) \hat{g} = h^{-1} x g$.

Hence $\hat{G}$ and $\check{G}$ commute and $\langle \hat{G}, \check{G} \rangle = \hat{G} \check{G}$

Moreover, $\hat{G} \check{G} = \hat{G} \circ \check{G}$ since $\hat{G}$ and $\check{G}$ commute.

($\varphi: M \times N \rightarrow MN$ is a homomorphism since $\hat{G}$ and $\check{G}$ commute.

$$(m, n) \mapsto mn$$

$$\ker \varphi = \{(\hat{g}, \check{g}) \mid g \in Z(G)\}, \quad \hat{g} \in Z(M)$$

$$\textcircled{3} \text{ Let } X = \langle \hat{G}, \check{G} \rangle = \hat{G} \circ \check{G}$$

$$\text{Then } X_w = \{(\hat{g}, \check{g}) \mid g \in G\} \quad \text{where } w = 1 \in G.$$

$$x(\hat{g}, \check{g}) = g^{-1} x g. \quad \hat{G} \text{ regular} \Rightarrow \hat{G} \cap X_w = 1 \text{ \& } \hat{G} \triangleleft X$$

$$\text{Thus } X_1 = \hat{G} \text{ and } X = \hat{G} : X_w (w=1) = \hat{G} : \hat{G} \text{ \& } X = \hat{G} X_w$$

$$\text{So } \hat{G} \check{G} = \hat{G} : \hat{G} = \check{G} : \hat{G} \quad (\text{内半直积}).$$

In particular, if $Z(G) = 1$, then $\hat{G} \times \check{G} = \hat{G} : \hat{G} = \check{G} : \hat{G}$.

Go back to case (1).

Identify Ω with M . Then $\hat{M}$ acts regularly on M by right multiplication.

Let $X = M \times N \leq \text{Sym}(\Omega)$.

$$\text{Then } X_w = \{(g, g') \mid g \in M\}$$

$$\varphi: M \rightarrow N \text{ s.t. } g^\varphi = g'$$

$$\Rightarrow X = \hat{M} \times \check{M} \text{ on } M.$$

try to prove two actions are equivalent. (Recall $\hat{M} \cong M$)

$$\varphi: M \rightarrow N \text{ s.t. } g' = g^\varphi$$

$$(g, 1) \mapsto (1, g)$$

$$M \times \{1\} \rightarrow \{1\} \times M.$$

$$X = MN = M \times N = M \times M = \hat{M} \times \check{M} \Rightarrow M \cong \hat{M}, N \cong \check{M}$$

$Z(M) = 1$. $M < \text{Sym}(\Omega)$ regular.

$$\Rightarrow C_{\text{Sym}(\Omega)}(M) \geq N$$

$$\Rightarrow C_{\text{Sym}(\Omega)}(M) = N \text{ by lemma. (M transitive. } C_{\text{Sym}(\Omega)}(M) \text{ s.r.)}$$

$$\text{Similarly, } C_{\text{Sym}(M)}(\hat{M}) = \check{M}$$

lemma

Let $M < \text{Sym}(\Omega)$ be regular. Then identifying Ω with M , the action of M on Ω is equivalent to the action of $\hat{M}$ on M .

$$\text{Pf: } \begin{cases} \Omega = \{w_1, w_2, \dots, w_n\} & w_i = w_i^{g_i} \end{cases}$$

$$M = \{1 = g_1, g_2, \dots, g_n\}$$

$$w_i^g = w_i^{g_i g} \mapsto g_i g$$

$$w_i \mapsto g_i, (g_i)^g = g_i g$$

$\Rightarrow$ equivalent

□

$$\text{Moreover, } N = C_{\text{Sym}(\Omega)}(M) = C_{\text{Sym}(M)}(\hat{M}) = \check{M}$$

Week 8 holomorph.

Corollary

Each regular action of a group X is equivalent to $\hat{X}$ acting on X / $\check{X}$ acting on X .

pf: we only prove for $\check{X}$

let $\Phi: X \rightarrow \Omega$

$$\varphi: \check{X} \rightarrow X$$

$$x_i \mapsto w^{x_i^{-1}} x_j \mapsto x_j$$

$$\Rightarrow x_i \check{x}_j = x_j^{-1} x_i \rightarrow w^{x_i^{-1}} x_j = (w^{x_i^{-1}}) x_j$$

Then Φ is 1-1 and ρ is iso.

hence two actions are equivalent.

Def holomorph (全形).

Let G be a group. The holomorph of G is $\text{Hol}(G) = G : \text{Aut}(G)$

$$(\text{Hol}(G) = \{ (g, \phi) \mid g \in G, \phi \in \text{Aut}(G) \} \text{ where } (g_1, \phi_1)(g_2, \phi_2) = (g_1 g_2^{\phi_1^{-1}}, \phi_1 \phi_2))$$

Case (1) holomorph simple.

$$M \times N \triangleleft G \leq \text{Sym}(\Omega) \text{ where } M \cong N \cong T \text{ simple}$$

$$\text{Then } M : \text{Inn}(M) \leq G \leq M : \text{Aut}(M)$$

pf:

$$\text{LHS: } \hat{M} \times \check{M} = M \times N \triangleleft G$$

$$\hat{M} = \check{M}$$

$$M : \text{Inn}(M)$$

$$\text{RHS: } G = M : G_0 \leq \text{Sym}(\Omega)$$

identify M with Ω , we have

Campus

$$G = \hat{M} : G_1 \leq \text{Sym}(M)$$

Claim: $G_1 \leq \text{Aut}(M)$.

$\forall g \in G_1, g \hat{m} g = \hat{m}'$ for some $m' \in M$.

Then $g^{-1} \hat{m} g (1) = \hat{m}'(1) \Rightarrow g(m) = m'$

Let $g^{-1} \hat{m} g = \hat{m}'$, $g^{-1} \hat{n} g = \hat{n}'$. Then $g^{-1} \hat{m} \hat{n} g = \hat{m}' \hat{n}'$ which yields $g(mn) = m'n' = g(m)g(n)$. Hence $G_1 \leq \text{Aut}(M)$.

Lemma (Frattini argument)

Let $H < G \leq \text{Sym}(\Omega)$ s.t. G is transitive on Ω .

Then H transitive on $\Omega \Leftrightarrow G = H G_w$ (i.e. $\forall g \in G, \exists x \in H, y \in G_w$ s.t. $g = xy$).

Lemma Let $M < G \leq \text{Sym}(\Omega)$ s.t. M is regular on G .

Then $G = M : G_w$ where $w \in \Omega$.

Case (2) holomorph compound.

$M \times N < G \leq \text{Sym}(\Omega)$

where $M \cong N = T^l$ with $l \geq 2$ and T nonabelian simple

Then $M : \tilde{M} \leq G \leq M : \text{Aut}(M)$ where $\text{Aut}(M) =$

$\text{Aut}(T) \wr_l S_l$ (wreath product).

$M \cong N = T^l = T_1 \times \dots \times T_l$ where $T \cong T_1 \cong \dots \cong T_l$

G induces a transitive action on $\{T_1, \dots, T_l\}$.

Lemma

A quasiprimitive permutation group of type HS, HC, or HA is primitive.

Let G be a transitive permutation group on Ω .

Def block & block system.

① A subset $B \subset \Omega$ is called a block if for any element $g \in G$, $B^g = \{w^g \mid w \in B\}$ either $B^g = B$ or $B^g \cap B = \emptyset$.

② For a block B , the set $\mathcal{B} = \{B^g \mid g \in G\}$ is called a block system for G on Ω .

Properties of blocks

Write $\mathcal{B} = \{B_1, B_2, \dots, B_m\}$.

① each B_i is a block.

② $B_i \cap B_j = \emptyset \quad \forall i \neq j$.

③ $|B_1| = \dots = |B_m|$

④ G induces a transitive action on $\mathcal{B}$.

Def primitive & imprimitive

If there is a block B for G on Ω with $1 < |B| < |\Omega|$ (i.e. nontrivial block B). Then G is said to be imprimitive on Ω . Otherwise G is said to be primitive.

Lemma A transitive group $G \leq \text{Sym}(\Omega)$ is primitive $\Leftrightarrow G_w$ is a maximal subgroup of G .

pf:

$\Leftarrow$ Assume otherwise G is imprimitive on Ω , let B be a nontrivial block. Let $w \in B$, then

$$G_w \not\subseteq G_B = \{g \in G, B^g = B\}.$$

($\exists w' \neq w \in B, \exists g \in G$ s.t. $w^g = w'$ then $g \in G_B$ but not G_w)

Campus

i.e. $G_w \not\subseteq G_B \not\subseteq G$

$\Rightarrow$ If $G_w < H < G$. then $B := w^H$ is a block.

① $B^g \cap B = \emptyset$ or $B^g = B$.

let $g \in G$. Consider $B^g \cap B$. Now suppose $s \in B^g \cap B$. Then $w^{h_1 g} = w^{h_2}$ for some $h_1, h_2 \in H$ so $h_1 h_2^{-1} g \in G_w < H$. Since $h_1, h_2 \in H, g \in H$, hence $B^g = B$.

② B is a nontrivial block.

use lemma. if H transitive, then $G = H G_w = H$. Hence H not transitive and $|B| < |\Omega|$. $\square$

pf of lemma: Case HA.

let G be of type HA. Then $G = M : G_w$ where M is ^{regular} abelian unique minimal normal subgroup.

Suppose G is imprimitive on $\Omega = M$. Then $\exists H$ s.t. $G_w < H < G$ and $1 \neq H \cap M = M_0 < M$. (if $H \cap M = 1$, then $G \leq M \cdot H$).

Then $M_0 \triangleleft M$, $M_0 \triangleleft H$, $M_0 \triangleleft \langle H, M \rangle = G$, contradiction!
 $\downarrow$
 M abelian

Counterexample example of being quasi-primitive but not primitive
 (note that primitive $\rightarrow$ quasiprimitive).

pf: let $H \triangleleft G$

① $H < G_w$ for some $w \in \Omega$.

Then $H < G_w^x = G_w^x$ for any $x \in G$.

i.e. H fixes any element in Ω since G is transitive $\Rightarrow H = 1$ since G faithful.

② $H \not\leq G_w$ for any $w \in \Omega$.

Then $H G_w$ is a group strictly larger than $G_w \Rightarrow G = H G_w$.

$(h_1 g_1 (h_2 g_2)^{-1}) = h_1 g_1 g_2^{-1} h_2^{-1} = h_1 g_1 g_2^{-1} h_2^{-1} g_2 g_1^{-1} g_1 g_2^{-1} \in H G_w \Rightarrow H$ transitive.
 $e \in H G_w$ $\forall h_1, h_2 \in H, g_1, g_2 \in G_w$

Let G be nonabelian simple.

Then $\hat{G}$ is a quasi-primitive group on G , but $\hat{G}$ is not primitive since $\overline{G} = 1$ is not maximal in $\hat{G}$.

Week 9HS type (HC type * (homework)).Let $G = T : H$ with $\text{Inn}(T) \leq H \leq \text{Aut}(T)$

- H can be G_w with $w=1 \in T$ (or any other w).
- Claim: G is primitive on $\Omega (\equiv T)$ ($\Leftrightarrow G_w$ is maximal)

pf: Suppose $H \leq K \leq G$.Then an elt $g \in K \setminus H$ has the form $g = (t, h) \in G = T : H$ such that $t \neq 1$. Thus $(t, 1) = (t, h)(1, h^{-1}) \in K$.For any elt $x \in H$, we have $(1, x^{-1}) \in K$ and $(1, x^{-1})(t, 1) \in K$.
 $(t^x, x^{-1})(1, x) = (t^x, 1) \in K$.
 (t^x, x^{-1}) Let $M = \langle (t^x, 1) \mid x \in H \rangle \leq T \times \{1\}$.Since $H \geq \text{Inn}(T)$, the action of T on M by conjugation fixes M , so $M \triangleleft T$. Since T simple, $\langle M, H \rangle = \langle T, H \rangle = T : H = G \leq K$. So H is a maximal subgroup of G and G is primitive on Ω .Corollary This provides a way to construct primitive groups. For T simple, let $G = T \times T = T : \tilde{T} = T : \text{Inn}(T)$ (since T simple, $Z(T)$ is trivial, hence semidirect product).NotationLet $G \leq \text{Sym}(\Omega)$, $|\Omega| = n$ — For $\Delta \subseteq \Omega$ and $g \in G$, let $\Delta^g = \{s^g \mid s \in \Delta\}$.— For $w_1, w_2 \in \Omega$, write $w_1 \sim w_2$ if $w_1^g = w_2$ for some $g \in G$.— $w^G = \{w^g \mid g \in G\}$ orbit of w .— $G_w = \{g \in G \mid w^g = w\}$ stabilizer of w .

— For $\Delta \subset \Omega$, let $G_\Delta = \{g \in G \mid \Delta^g = \Delta\}$ ($\cong G_{\{\Delta\}}$)
 $G_{(\Delta)} = \{g \in G \mid w^g = w \ \forall w \in \Delta\}$.

$\Rightarrow G_{(\Delta)} \triangleleft G_\Delta$ and G_Δ acts on Δ with kernel $= G_{(\Delta)}$

— $g^{-1} G w g = G w^g$

$$g^{-1} G_\Delta g = G_{\Delta^g}$$

$$g^{-1} G_{(\Delta)} g = G_{(\Delta^g)}$$

— G_Δ acting on Δ induces a permutation group on Δ , denoted by G_Δ^Δ and $G_\Delta^\Delta \cong G_\Delta / G_{(\Delta)}$.

— $\text{Fix}_\Omega(G) = \{w \in \Omega \mid w^g = w \ \forall g \in G\}$.

The set of fixed points of G on Ω .

— $\text{Fix}_\Omega(g) = \{w \in \Omega \mid w^g = w \ \forall g \in G\}$.

Def semiregular

G is said to be semiregular on Ω if $G_w = \{1\}$ for each point $w \in \Omega$.

- $|G| = \text{length of each orbit} \Rightarrow |G| \mid |\Omega|$ and orbit length $\mid |\Omega|$
- If G is transitive and semiregular, then G is regular on Ω and $|G| = |\Omega|$

Def coset action. (G transitive on Ω).

Let G be a group and $H \leq G$. Let $\Omega = [G:H] = \{Hg \mid g \in G\}$. Define an action of G on Ω by

$$(Hg)^x = Hgx \in [G:H] \quad \forall Hg \in \Omega \text{ and } x \in G$$

i.e. $x: Hg \mapsto Hgx$

• A coset action is transitive ($(Hg_1)^{g_1^{-1}g_2} = Hg_2$).

• The kernel of this action is $G_{(\Omega)} = \bigcap_{g \in G} g^{-1} H g$.

Denote G cos by M . Then $M = \bigcap_{g \in G} g^{-1} H g \triangleleft G$, $M \leq H$ and M is the largest normal subgroup of G contained in H called the core of H in G denoted by $\text{Core}_G(H)$.

prop $G/\text{Core}_G(H) \cong G^\Omega$ is a transitive permutation group on $\Omega = [G:H]$.

• If $\text{Core}_G(H) = 1$, then the action is said to be faithful (H is core-free in G). Otherwise the action is unfaithful.

Thm

Each transitive permutation group is equivalent to a faithful coset action.

pf:

let G be a transitive permutation group on Ω i.e. $G \leq \text{Sym}(\Omega)$ is transitive.

let $H = G_w$ with $w \in \Omega$. let $\Sigma = [G:H]$. Then for $G = Hg$ and $x \in G$, $G^x = (Hg)^x = Hgx \in \Sigma$.

$\Omega: w, w^{g_1}, w^{g_2}, \dots, w^{g_n}$

$\Sigma: H, Hg_1, Hg_2, \dots, Hg_n$

For $x \in G$, s.t. $x: w^{g_i} \rightarrow w^{g_j}$
 $Hg_i \rightarrow Hg_j$

• H is core-free in G i.e. $\text{Core}_G(H) = 1$ i.e. H does not contain a nonidentity normal subgroup of G .

pf: Suppose $N \leq H$ and $N \triangleleft G$.

Then $N = Nw$ since $N \leq H$. for $x \in G$, $Nw = N = x^{-1}Nx$
 $= x^{-1}Nwx = Nw^x$ so N fixes every point of Ω . KOKUYO

We then must have $N=1$ as $G \leq \text{Sym}(\Omega)$. Thus $\text{Core}_G(H) = 1$ and G acts on Σ faithfully.

Def equivalence of action. (same G)

Let G acts on Ω and Σ transitively and faithfully. Then G^Ω and G^Σ are said to be equivalent if there is a H map φ from Ω to Σ s.t. for any $w \in \Omega$,

$$\begin{array}{ccc} w & \xrightarrow{\varphi} & w^\varphi = \sigma \\ g \downarrow & & \downarrow g \\ w^g & \xrightarrow{\varphi} & (w^g)^\varphi = (w^\varphi)^g \end{array}$$

Example equivalent or not

$G = S_4$ on $\{1, 2, 3, 4\}$

$$H_1 = \langle (12) \rangle$$

$$H_2 = \langle (12)(34) \rangle$$

$$H_3 = \langle (13) \rangle$$

$$H_4 = \langle (13)(24) \rangle$$

$$\Rightarrow \Omega_i = [G : H_i] \Rightarrow G^{\Omega_1} \text{ and } G^{\Omega_2} \quad \times$$

$$G^{\Omega_1} \text{ and } G^{\Omega_3} \quad \checkmark$$

Pf:

① G^{Ω_1} and G^{Ω_2} : note that $H_1(12) = H_1$

If $H_1 \rightarrow H_2 g$ for some $g \in G$. then we must have

$$H_2 g = H_2 g(12) \text{ which means}$$

$$1^\circ \begin{cases} g(12) = g \end{cases}$$

$$2^\circ \begin{cases} g(12) = (12)(34)g \end{cases}$$

$$\begin{cases} (12)(34)g(12) = (12)(34)g \end{cases}$$

$$\begin{cases} (12)(34)g(12) = g \end{cases}$$

both not possible.

② G^{Ω_1} and G^{Ω_2}

By above 1° and 2° first row is some conjugation relation.

then we may take $H_1 \rightarrow H_3(23)$.

$$\text{So } \varphi: [G: H_1] \rightarrow [G: H_3]$$

$$H_1 \quad H_1 g$$

$$H_3(23) \quad H_3(23)g$$

Only need to check φ is well-defined.

Indeed, if $H_1 g_1 = H_1 g_2$ then $g_1 g_2^{-1} \in H_1 = H_3^{(23)}$

$$\text{Hence } (23)H_3(23)g_1 = (23)H_3(23)g_2 \Rightarrow H_3(23)g_1 = H_3(23)g_2$$

Thm criterion for equivalence

G^Ω and G^Σ both transitive are equivalent



G_Ω and G_Σ are conjugated in G .

Pf:

$\Leftarrow$ defined map is well-defined

$$\Rightarrow \Omega \rightarrow \Sigma : g G_\Omega g \subset G_\Sigma$$

$$\Sigma \rightarrow \Omega : g' G_\Sigma g' \subset G_\Omega$$

Week 10.

Exercise 13

Characterize quasiprimitive permutation group $G \leq \text{Sym}(\Omega)$ such that G_ω has r orbits on Ω where $r=2, 3, 4, 5$.
If G is of holomorph simple type, then $r \geq ?$

Thm (Burnside)

A 2-transitive permutation group is affine (holomorph affine) or almost simple (AS) $\downarrow \text{rank}(G)=2$.

Def $\text{rank}(G)$

Let $G \leq \text{Sym}(\Omega)$ be transitive. Then G induces an action on $\Omega \times \Omega = \{(w_1, w_2) \mid w_1, w_2 \in \Omega\}$ by
 $(w_1, w_2)^g = (w_1^g, w_2^g)$ $\downarrow$ ordered.

Then G acts intransitively and partitions $\Omega \times \Omega$ into G -orbits. The number of G -orbits is called the rank of G on Ω , denoted by $\text{rank}(G)$.

• $\{(w, w) \mid w \in \Omega\}$ is an orbit. so $\text{rank}(G) \geq 2$

If $\text{rank}(G)=2$, then G is called 2-transitive.

• A 2-trans group is primitive.

• If G is 2-transitive on Ω . Then

i) G_ω is faithful and transitive on $\Omega \setminus \{\omega\}$.

ii) $|\Omega| - 1$ divides $|G_\omega|$

iii) If $N \triangleleft G$, then N_ω is half-transitive on $\Omega \setminus \{\omega\}$.
 $\downarrow$
 orbits have the same size.

pf: (iii) use the following lemma with

Campus

$N_\omega = N \cap G_\omega \triangleleft G_\omega$ where G_ω is transitive on $\Omega \setminus \{\omega\}$.

Lemma G transitive, $N \triangleleft G \Rightarrow N$ half-transitive.

Let $G \leq \text{Sym}(\Omega)$ be transitive. $1 \neq N \triangleleft G$ intransitive. Then the N -orbits on Ω form a nontrivial block system and N is half-transitive.

pf: Let Δ_1, Δ_2 be two N -orbits. $s_i \in \Delta_i$ with $i=1$ or 2 .
Then $s_1^g = s_2$ for some $g \in G$.

Write $\Delta_1 = \{s_1, s_1^{x_2}, \dots, s_1^{x_t}\}$ with $x_i \in N$.

Then $(s_1^{x_i})^g = s_1 g g^{-1} x_i g = (s_1^g)^{g^{-1} x_i g} = s_2^{g^{-1} x_i g} \in \Delta_2$.

So $\Delta_1^g \subseteq \Delta_2$. Similarly $\Delta_2^g \subseteq \Delta_1$, so $|\Delta_1| = |\Delta_2|$.

Corollary If G is primitive, then G is quasiprimitive.

pf of thm

Let $G \leq \text{Sym}(\Omega)$ be 2-transitive. Let $N \triangleleft G$ s.t. $N = T^k$ with $k \geq 2$ and T nonabelian simple.

Lemma 1

N does not have a proper normal subgroup which is regular on Ω (i.e. G is not HS, HC, SD, CD).

pf:

Suppose $M \triangleleft N$ is regular on Ω . Then $|\Omega| = |M|$ and $N_w \neq 1$ (since $M \triangleleft N$) is half-transitive on $\Omega \setminus \{w\}$.

Let m be the size of an orbit of N_w on $\Omega \setminus \{w\}$.

Then $m \mid |\Omega \setminus \{w\}|$ and $m \mid |N_w|$. Thus $m \mid \gcd(|\Omega| - 1, |N_w|)$. Now $|\Omega| = |M| = |T|^k$ for $k \geq 2$ and $N = M : N_w$ so that $|N_w| = |T|^{k-1}$. Thus $m \mid \gcd(|T|^k - 1, |T|^{k-1}) = 1 \Rightarrow m = 1$ contradiction. (N_w is faithful on $\Omega \setminus \{w\}$)

Lemma 2

N is not regular on Ω . i.e. G is not TW)

pf:

Suppose N is regular on Ω . Then $G = N G_w$ and $N \cap G_w = N_w = 1$ hence $G = N \cdot G_w$ and Ω can be identified with N with $w \rightarrow 1$ s.t. G_w acts on N by conjugation. ($w^n \mapsto n$; $wng = wng^{-1}ng = wng^{-1}ng \mapsto g^{-1}ng = ng^{-1}$).

The action of G_w partitions N into orbits and fixes 1.

Note that elts of N in the same orbits are of same order.

Clearly $|N|$ has at least 3 prime divisors r, s, t .

($N = T^k$ where $k \geq 2$, T nonabelian simple is not solvable while groups of odd order are solvable, groups of order $p^a q^b$ are solvable by Burnside). Then $\exists x, y, z \in N$ s.t. $|x| = r, |y| = s, |z| = t$. Thus $x^{G_w}, y^{G_w}, z^{G_w}$ are 3 distinct orbits of G_w on N . Since $\text{rank}(G) = \#$ of G_w orbits on Ω , we have $\text{rank}(G) \geq 4$. a contradiction.

Remark example of " $= 4$ " holds: If $N = T = A_5$, then $(r, s, t) = 2, 3, 5$. $G_w = S_5$ $\text{rank}(G) = 4$. (HS).

Lemma 3

Let G be 2-transitive and let $N \triangleleft_{\min} G$ such that $N = T^k$ does not have a normal subgroup which is regular on Ω with $k \geq 2$ and T nonabelian simple ($\Rightarrow$ Now N is the unique minimal normal subgroup). Then $\Omega = \Delta^k$ s.t. G acts on Ω by $(\delta_1, \dots, \delta_k)^{(z_1, \dots, z_k)\pi} = (\delta_{1, \pi^{-1}}, \dots, \delta_{k, \pi^{-1}})$ (why we call type PA (G acts on Ω in product action)).

pf: $N_w \neq 1$. Write $N = T_1 \times \dots \times T_k$ where $T_1 \cong \dots \cong T_k$.
 let H_i be the projection image of N_w into T_i ($g \in N_w \mapsto (t_1, \dots, t_k)$ where $t_i \in T_i \mapsto t_i$ or $(1, 1, \dots, t_i, 1, \dots, 1)$). Let $M_i = \prod_{j \neq i} T_j$. $M_i \triangleleft N$ and $N/M_i \cong T_i$.
 Then $H_i = N_w / (N_w \cap M_i) = N_w M_i / M_i$ and $H_i \leq T_i$ with $H_1 \cong H_2 \cong \dots \cong H_k$ if not, any orbit is a proper smaller normal subgroup ($G = N G_w$ transitive on $\{T_1, \dots, T_k\}$ ($T_i^g \triangleleft N \Rightarrow T_i^g = T_j$ for some $j \forall g \in G$) and N trivial on $\{T_1, \dots, T_k\}$ by conjugation so G_w transitive on $\{T_1, T_2, \dots, T_k\}$. Also, G_w normalizes N_w so G_w is transitive on $\{H_1, \dots, H_m\}$ by conjugation).

Then $N_w \leq \underbrace{H_1 \times \dots \times H_k}_K \leq N$ and K is normalized by G_w .
 $\downarrow N$ transitive.

Thus $G_w \leq K G_w < G$. Since G primitive, G_w maximal.
 Then $G_w = K G_w \Rightarrow K = N_w \leq G_w$ ($K \leq N$ and $K \leq G_w$).
 $\uparrow K G_w \neq G$. otherwise $K < G$ But $K < N$.

Consider $N = T_1 \times \dots \times T_k$ acts on Ω .

Identify Ω with $[N : N_w]$

$$\begin{aligned} \Omega &= [N : N_w] = [T_1 \times \dots \times T_k : H_1 \times \dots \times H_k] \\ &= [T_1 : H_1] \times \dots \times [T_k : H_k] \\ &\quad \underbrace{\quad}_{\Delta_1} \qquad \qquad \qquad \underbrace{\quad}_{\Delta_k} \end{aligned}$$

with N acting on Ω defined by
 $w^{(t_1, \dots, t_k)} = (H_1 x_1, \dots, H_k x_k)^{(t_1, \dots, t_k)}$
 $= (H_1 x_1 t_1, \dots, H_k x_k t_k)$

$G_w \leq (L_1 \times \dots \times L_k) : S_k$ where $H_i \leq L_i \leq \text{Aut}(T_i)$

$G \leq (\text{Aut}(T_1) \times \dots \times \text{Aut}(T_k)) : S_k = (T_1 \times \dots \times T_k) : (\text{Aut}(T_1) \times \dots \times \text{Aut}(T_k))$
 $(G(N) = 1)$.

$$\begin{aligned}
 g &= (z_1, \dots, z_k) \pi \text{ where } T_i \in \text{Aut}(T_i), \pi \in S_k. \\
 \Rightarrow w &= (\delta_1, \dots, \delta_k) \in \Delta_1 \times \dots \times \Delta_k, \quad g = (z_1, \dots, z_k) \pi. \\
 \Rightarrow w^g &= (\delta_1, \dots, \delta_k) (z_1, \dots, z_k) \pi \\
 &= (\delta_1^{z_1}, \dots, \delta_k^{z_k}) \pi \\
 &= (\delta_{1, \pi^{-1}}^{z_1 \pi^{-1}}, \dots, \delta_{k, \pi^{-1}}^{z_k \pi^{-1}}) \in \Delta^k
 \end{aligned}$$

□.

Example

$$\begin{aligned}
 G &= (\text{Aut}(T) \times \text{Aut}(T)) : S_2 \quad \rightarrow \text{can also be } (T \times T) : S_2 \\
 &\quad \text{i.e. } (A_5 \times A_5) : S_2 \\
 \Omega &= \Delta_1 \times \Delta_2 = \Delta^2 \quad \Omega = \Delta^2 \\
 &\quad |\Delta| = 5. \\
 G_w &= (L_1 \times L_2) : S_2 \quad (H_i \leq L_i \leq \text{Aut}(T_i)) \quad G_w = (A_4 \times A_4) : S_2
 \end{aligned}$$

e.g.

$$\begin{aligned}
 \Delta &= \{0, 1, 2, 3, 4\}. \\
 \Omega &= \Delta^2 = \{(i, j) \mid 0 \leq i, j \leq 4\}. \\
 \text{Fix } w &= (0, 0). \quad G_w = (\text{Alt}(1, 2, 3, 4) \times \text{Alt}(1, 2, 3, 4)) : \pi \\
 &\quad (\pi : (t_1, t_2) \rightarrow (t_2, t_1)) \\
 \text{Then } G_w &\text{ partitions } \Omega \setminus \{w\} = \Delta^2 \setminus \{(0, 0)\} \text{ into two orbits,} \\
 &\{ (0, i), (i, 0) \mid 1 \leq i \leq 4 \} \\
 &\{ (i, j) \mid 1 \leq i, j \leq 4 \}. \\
 (0, 1) &\xrightarrow{(1, 2)(3, 4)} (0, 2); \quad (0, 1) \xrightarrow{\pi} (1, 0); \quad (1, 1) \xrightarrow{(1, i)(k, k_2), (j)(k'_1 k'_2)} (i, j). \\
 \text{so rank}(G) &= 3.
 \end{aligned}$$

Lemma (HW)

$G = (T \times T) : S_2$ acting on $\Omega = \Delta^2$ is of rank at least 3 and $\text{rank}(G) = 3 \iff T$ is 2-transitive on Δ

problem

Let $G = (T \times T \times T) : S_3$ acts on $\Omega = \Delta^3$ in product action.
Find a sharp lower bound for $\text{rank}(G)$ ("=" may hold)
 $\text{rank}(G) \geq 4 \iff T \text{ 2-transitive.}$

(clue: $\Delta = \{0, 1, \dots, n-1\}$

$$\Omega = \Delta^k = \{(\delta_1, \dots, \delta_k)\}.$$

$$w = (0, \dots, 0).$$

If $w \in \Omega$, let $\text{weight}(w) = \# \text{ of } \delta_i \neq 0$.

Then same weight $\Rightarrow$ same orbit).

Week 11

$$\text{rank}(G) = 3$$

Thm If $\text{rank}(G) = 3$ and G is quasi-primitive on Ω , then G is HA, AS or $N = T^2$ and G acts on $\Omega = \Delta^2$ in product action with T on Δ is 2-transitive.

Counter example $G = A_5 : S_5 = \text{Hol}(A_5)$ rank (4).
"Aut(A_5).

(A_5 acts on A_5 by right multiplication.

$S_5 = G_w$ acts on A_5 by conjugation. (proved below in lemma 1)

identify w with 1

Then orbits of G is conjugacy classes of A_5 under S_5 :
 $O_1 = \{1\}$, $O_2 = \{(ij)(kl)\}$, $O_3 = \{(ijk)\}$, $O_4 = \{(ijklm)\}$.

Lemma 1

Let $M \trianglelefteq G$ be regular on Ω . Then $\text{rank}(G) \geq$
of fusion classes of elts of M .

↓ orbits under $\text{Aut}(M)$

In particular, if $M = T^k$ with T nonabelian simple, then $\text{rank}(G) \geq 4$.

($\Rightarrow G$ is not of type HS, HC, TW).

pf:

Identify Ω with M . let $w \equiv 1$. Then $G = M : G_w$ and G_w acts on $(M \equiv \Omega)$ by conjugation...

(G acts on $\Omega = [G : G_w] = \{G_w g \mid g \in G\}$

$$h \in G_w : G_w g \xrightarrow{h} = G_w g h = G_w h^{-1} g h = G_w g^h$$

let g be in $M \Rightarrow g^h \in M$).

Now the action of G_w on M partitions into orbits i.e. fusion classes of M since $G_w \leq \text{Aut}(M)$. So $\text{rank}(G) \geq$ # of fusion classes of M since $G_w \leq \text{Aut}(M)$.

classes of M .

Suppose $M = T^k$ with T nonabelian simple $\Rightarrow M$ not solvable ($|M| \neq p^a q^b$). Then there exists distinct primes r, s, t dividing $|M|$. Let $x, y, z \in M$ s.t. $|x| = r, |y| = s, |z| = t$. Then $x^{G_M}, y^{G_M}, z^{G_M}$ are three different fusion classes of M . so $\text{rank}(G) \geq 4$. $\square$

lemma 2

Let G be a quasi-primitive of type SD. (simple diagonal). Then $\text{rank}(G) \geq 4$.

pf: Let $N = T^k \triangleleft G$ and write $N = T_1 \times \dots \times T_k$ where $T_1 \cong \dots \cong T_k \cong T$. $k \geq 2$. $\Rightarrow$ each reordering T_i is an iso by properties of N_w .
 WLOG $N_w = \{(t, \dots, t) \mid t \in T\} \cong T$ and $R := T_1 \times \dots \times T_{k-1} \times T_k$.
 unique elt in N_w is $(t_1, \dots, t_k)$. then t_i defines a unique elt in N_w .

R is regular on Ω . ($N = RN_w$ is transitive $\rightarrow R$ transitive).

Identify Ω with $[N: N_w] = \{N_w(t_1, \dots, t_k) \mid t_i \in T\}$.

where $N_w(t_1, \dots, t_k) = \{(st_1, \dots, st_k) \mid s \in T\}$ with a representative $(t_k^{-1}t_1, \dots, t_k^{-1}t_{k-1}, 1) \in R$.

Observe $N_w(s_1, \dots, s_{k-1}, 1) = N_w(t_1, \dots, t_{k-1}, 1) \Leftrightarrow (s_1, \dots, s_{k-1}, 1) = (t_1, \dots, t_{k-1}, 1) ((s_1^{-1}t_1, \dots, s_{k-1}^{-1}t_{k-1}, 1) \in N_w)$.

Now, the stabilizer G_w satisfies $N_w \cdot P \leq G_w \leq \text{Aut}(T) \times S_k$ where $P \leq S_k$ is transitive. (G_w acts by conjugation; $N_w \leq G_w$).

G_w acts on T_i transitively and acts faithfully on N as

$C_G(N) \triangleleft G$ and $C_G(N) \cap N = \{1\} \Rightarrow C_G(N) = \{1\}$. $G_w \leq \text{Aut}(N)$.

$\times S_k$ as G_w stabilizes N_w and $\text{Aut}(T) \times S_k$ is a direct product as $\text{Aut}(T) \times \dots \times \text{Aut}(T)$ becomes $\text{Aut}(T)$ thus S_k acts on $\text{Aut}(T)$ trivially.

Each element of G_w has the form 6π with $6 \in \text{Aut}(T)$ and $\pi \in S_k$ and an elt $(t_1, \dots, t_n) \in N$ is sent by 6π to $(t_1, \dots, t_k)^{6\pi} = (t_1^6, \dots, t_k^6)^\pi \in N$.

Pick two elts $x := (s_1, 1, \dots, 1), y := (t_1, 1, \dots, 1) \in R \setminus \{(1, \dots, 1)\}$.

Then x, y lie in the same orbit of G_w iff

$\exists 6\pi \in G_w$ s.t.

$$(s_1, 1, \dots, 1) = (t_1, 1, \dots, 1)^{6\pi} = (t_1^6, 1, \dots, 1)^\pi$$

° If $1^\pi = 1$, then $(s_1, 1, \dots, 1) = (t_1^6, 1, \dots, 1)$ so $s_1 = t_1^6$

i.e. s_1, t_1 are conjugated by $6 \in \text{Aut}(T)$.

° If $1^\pi = k$, then $(s_1, 1, \dots, 1) = (1, \dots, 1, t_1^6)$

$$= ((t_1^6)^{-1}, \dots, (t_1^6)^{-1}, 1) \in R.$$

So $1 \neq s_1 = (t_1^6)^{-1} = (t_1^{-1})^6$ and $k=2$ (otherwise $(t_1^6)^{-1} = 1$ impossible)

3° If $1^\pi = \{1, k\}$, say $1^\pi = 2$ WLOG.

Then $(s_1, 1, \dots, 1) = (1, t_1^6, 1, \dots, 1) \in R$ so $s_1 = 1$ contradiction.

Therefore $(s_1, 1, \dots, 1)$ and $(t_1, 1, \dots, 1)$ lie in the same orbit of $G_w \Leftrightarrow$ either $s_1 = t_1^6$ or $k=2$ and $s_1 = (t_1^{-1})^6$

Thus $\text{rank}(G) \geq \#$ of orbits of $\text{Aut}(T)$ on $\{(t, t^{-1}) \mid t \in T\}$. $\downarrow$ s_1 and t_1 are inverse-conj under $\text{aut}(T)$.

≥ 4

(3 distinct primes divide order of the group. two elts of distinct order are not conj. nor inverse conj. in auto group).

Example $G = (A_5 \times A_5) \cdot \begin{matrix} \langle 6 \rangle & \langle \pi \rangle \\ \downarrow & \text{out}(T) \\ \langle 2 \times S_2 \rangle \end{matrix} \Rightarrow G_w = S_5 \times S_2 = \text{Aut}(T) \times S_2$
rank = 4.

smaller G can be $G = A_5^2 : \left. \begin{matrix} \langle 6 \rangle : \langle \pi \rangle \\ \langle 6\pi \rangle \\ \langle \pi \rangle \end{matrix} \right\} \text{ both transitive. (has } S_2).$

Def wreath product

$$G = T \wr S_k = T^k : S_k \text{ wreath product.}$$

where the semiproduct is defined as

$$(t_1, t_2, \dots, t_k)^\pi = (t_1 \pi^{-1}, t_2 \pi^{-1}, \dots, t_k \pi^{-1})$$

or (equivalently) by multiplication of two elements

$$\begin{aligned} & (s_1, \dots, s_k) \pi_1 (t_1, t_2, \dots, t_k) \pi_2 \\ &= (s_1, \dots, s_k) \pi_1 (t_1, \dots, t_k) \pi_1^{-1} \pi_1 \pi_2 \\ &= (s_1 t_1^{\pi_1}, s_2 t_2^{\pi_1}, \dots, s_k t_k^{\pi_1}). \end{aligned}$$

Def product action.

Let T be a transitive permutation group on $\Delta = \{0, 1, \dots, m\}$

For an integer $k \geq 2$, let $P \leq S_k$ be transitive, and let

$$G = T \wr P \leq T \wr S_k \text{ and } \Omega = \Delta^k$$

such that G acts on Ω naturally, namely

$$(s_1, \dots, s_k) (t_1, \dots, t_k)^\pi = (s_1 t_1^{\pi_1}, \dots, s_k t_k^{\pi_k})^\pi = (s_1 \pi_1^{-1}, \dots, s_k \pi_k^{-1})$$

In this way, G is said to be acting on $\Omega = \Delta^k$ in product action or G is a blow-up of T on Δ .

Fix a point $w_0 = (0, \dots, 0)$. For a point $w = (s_1, \dots, s_k) \in \Omega$, define the weight of w :

$$\text{weight}(w) = |\{i \mid s_i \neq 0\}|,$$

the number of nonzero entries of $w = (s_1, \dots, s_k)$. Then the action of each element $(t_1, \dots, t_k)^\pi \in N_{w_0} \xrightarrow{N \cong T^k} S_k$ preserves the weight of every point of every point of $\Omega \setminus \{w_0\}$. Thus w_1 not equivalent to w_2 under G_{w_0} if $\text{weight}(w_1) \neq \text{weight}(w_2)$.

Lemma 3 (Case product action).

Using the notation defined above. ($G = T \wr P \leq T \wr S_k$). We have

- (1) $\text{rank}(G) \geq k+1$
- (2) $\text{rank}(G) = 3$ iff $k=2$ and T is 2-transitive on Δ

pf:

- (1) The weight function ranges $\{0, 1, \dots, k\}$.

So G_{w_0} has at least $k+1$ orbits on Ω .

- (2) Assume $\text{rank}(G) = 3$. Then $k=2$ ($k \geq 2$). and $\Omega = \Delta \times \Delta$ and $G = (T \times T) : S_2$ with weight equal to 1 forming one orbit of G_{w_0} , so $\{(\delta_i, 0) \mid \delta_i \in \Delta \setminus \{0\}\}$ is an orbit of T_0 , i.e. T is 2-transitive on Δ . Converse is easy.

Week 12 Examples.

Let $V = F^d$ ($d > 1$) and let $G = GL(V)$ acts on V with orbits $\{0\}$, $V \setminus \{0\} = \Omega$

• G acting on Ω is transitive

(1) If $|F| > 2$, then the action is imprimitive.

pf: $\{ \langle v \rangle \mid v \in V \setminus \{0\} \}$ form a block system B .

($|\langle v \rangle| = |F| - 1$)

(2) If $|F| = 2$, then the action is 2-transitive.

pf:

Let $(u_1, u_2), (v_1, v_2) \in \Omega \times \Omega$, $u_1 \neq u_2$, $v_1 \neq v_2$.

Then u_1, u_2 are l.i. so is v_1 and v_2 . Then extend to bas

$u_1, \dots, u_d$

$v_1, \dots, v_d$

$\exists g \in GL(V)$ s.t. $u_i^g = v_i$

(3) If $|F| > 2$, $G = GL(V)$ acts on B transitively, with kernel equal to $Z(G)$. Then $G/Z(G) = PGL(V)$ is 2-transitive on B .

(4) $PGL(V)$ on B is not 3-transitive $\Leftrightarrow d > 2$

pf:

$\Leftarrow$ easy. Choose one set of l.i. and one set of l.d.

$\Rightarrow$ if $d = 2$.

Let $\langle u_1 \rangle, \langle u_2 \rangle, \langle u_3 \rangle$ be pairwise different, $\langle v_1 \rangle, \langle v_2 \rangle, \langle v_3 \rangle$ be pairwise different

Then u_1 and u_2 are l.i. and $u_3 = au_1 + bu_2$ with $a, b \in F$

Similarly, v_1 and v_2 are l.i. and $v_3 = cv_1 + dv_2$ with $c, d \in F$

Need to find $g \in GL(V)$ s.t. $(u_1, u_2)^g = (v_1, v_2)$. $u_3^g = (au_1 + bu_2)^g =$

$av_1 + bv_2$ $\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$ $cv_1 + dv_2$.

So $g: \begin{matrix} u_1 \rightarrow a^{-1}v_1 \\ u_2 \rightarrow b^{-1}v_2 \end{matrix}$ then $u_3 \rightarrow v_3$.

• Suppose $|F| > 2$. Then $GL(V)$ acting on $V \setminus \{0\} = \Omega$ is transitive and imprimitive, of rank $= 1 + 1 + |F^*| - 1 = |F|$ (F finite)

pf: rank $(G^\sim) = \#$ of G -orbits on $\Omega \times \Omega$

$$\{(u, v) \mid v \in \Omega\}$$

$$\{(u, v) \mid \langle u \rangle = \langle v \rangle, u, v \in \Omega\} \rightarrow (u, \lambda u) \rightarrow (v, \lambda' v)$$

$$\{(u, v) \mid \langle u \rangle \neq \langle v \rangle, u, v \in \Omega\}$$

$$\text{then } u^g = v$$

$$\lambda u^g = \lambda v$$

$$\lambda = \lambda'$$

$\Rightarrow$ partitions into $|F^*| - 1$ orbits

Let $N = \mathbb{Z}_p^d$ with p prime, $d \geq 1$

Then $H = \text{Aut}(N) = GL_d(p)$

Let $G = N \cdot H = \text{Hol}(N)$

Let $\Omega = N$ and define an action of $G = \mathbb{Z}_p^d : GL_d(p)$ on Ω :

$$w^{xy} = (w+x)^y = w^y + x^y \in N$$

$\downarrow$ operation in N is $+$

Claim: G is an affine 2-transitive permutation group on Ω .

Furthermore, if $p=2$, then G is 3-transitive (use (2) above).

pf: Let $(u_1, u_2), (v_1, v_2) \in \Omega \times \Omega$

need $xy = g \in N: H$ s.t.

$$(u_1, u_2)^{xy} = (v_1, v_2)$$

$\parallel$

$$((u_1+x)^y, (u_2+x)^y)$$

$$(u_1^y + x_1^y, u_2^y + x_2^y)$$

1° can just use G 2-transitive on $\Omega \Leftrightarrow G_w$ is transitive on

and G is transitive on Ω .

Remark: easy to see N is a minimal normal ^{No.} subgroup. 101
 every normal subgroup contains N , N is transitive on N . so G is quasi-primitive $\Rightarrow$ affine.

2° y maps l.d. to l.d. ; l.i. to l.i.

both l.i. : $\exists y \in G: H \checkmark$

u_1, u_2 l.i. & v_1, v_2 l.d. : let $x = au_1 + bu_2$.

$$u_1 + x = (1+a)u_1 + bu_2$$

$$u_2 + x = au_1 + (1+b)u_2 \quad (\alpha \neq 1).$$

$$\Rightarrow (1+a)u_1 + bu_2 = \lambda (au_1 + (1+b)u_2)$$

$$\Rightarrow 1+a = \lambda a \quad \Rightarrow a = \frac{1}{\lambda-1}$$

$$b = \lambda(1+b) \quad b = \frac{\lambda}{1-\lambda}$$

u_1, u_2 l.d. & v_1, v_2 l.i. : let x l.i. of u_1

then $u_1 + x$ and $u_2 + x$ are l.i.

u_1, u_2 l.d. & v_1, v_2 l.d. $u_1 = \lambda u_2 \quad v_1 = \lambda' v_2 \quad \lambda, \lambda' \neq 1$

$$\text{let } x = y u_2$$

$$\frac{x+u_1}{x+u_2} = \frac{\lambda+y}{1+y} = \lambda' \Rightarrow (x'-1)y = \lambda - \lambda'$$

$$y = \frac{\lambda - \lambda'}{\lambda' - 1}$$

Def k -transitive

G is said to be k -transitive on Ω if for any k -tuples $(w_1, \dots, w_k), (u_1, \dots, u_k)$ where w_i 's are distinct, u_i 's are distinct, then $\exists g \in G$ s.t. $(w_1, \dots, w_k)^g = (u_1, \dots, u_k)$
 i.e. $w_i^g = u_i$

prop G k -transitive $\Leftrightarrow G_w$ $k-1$ transitive on $\Omega \setminus \{w\}$ and G transitive on Ω .

pf:

$\Rightarrow$ easy

$\Leftarrow_\lambda (w_1, \dots, w_k)^g = (u_1, \dots, u_k)$ w_i 's are distinct, u_i 's are distinct
 need to find $g \in G$ s.t.

Since G transitive on Ω , $\exists g_1, g_2 \in G$ s.t. $w_1^{g_1} = w$, $u_1^{g_2} = w$.

$$\text{Then } (w_1, \dots, w_k)^{g_1} = (w, w_2^{g_1}, \dots, w_k^{g_1})$$

$$(u_1, \dots, u_k)^{g_2} = (w, u_2^{g_2}, \dots, u_k^{g_2})$$

$$\exists g \in G \text{ s.t. } (w_2^{g_1}, \dots, w_k^{g_1})^g = (u_2^{g_2}, \dots, u_k^{g_2})$$

$$\text{so } (w_1, \dots, w_k)^{g_1 g g_2^{-1}} = (u_1, \dots, u_k).$$

Question: when is sharply 2-transitive? (smaller group is no longer 2-transitive).

Def Sharply k -transitive.

G is said to be sharply k -transitive on Ω if for $(w_1, \dots, w_k), (w_1', w_2', \dots, w_k')$, there is exactly one elt $g \in G$ s.t. $(w_1, \dots, w_k)^g = (w_1', \dots, w_k')$.

Example

Let $F = \mathbb{F}_q$ where $q = p^f$

Let $F^+ \cong \mathbb{Z}_p^f$

$$F^+ \cong \mathbb{Z}_p^{f-1} = \langle \rho \rangle$$

Then $F^+ = \langle \rho \rangle$ acts on F^+ naturally by the field multiplication:

$$v \rho = v \rho \in F^+$$

ρ is an automorphism of the group F^+ , so $\rho \in \text{GL}(f, p)$.

and $G := F^+ : \langle \rho \rangle \leq \text{Hol}(F^+) = \mathbb{Z}_p^f : \text{GL}(f, p)$.

Note that $\langle \rho \rangle = \mathbb{Z}_p^{f-1}$ is transitive on $F^+ \setminus \{0\}$, i.e. for $u, v \in F^+ \setminus \{0\}$, $u = \rho^i, v = \rho^j$ then $u = v \rho^{i-j}$. So G is 2-transitive on $F^+ \setminus \{0\}$.

Counting numbers: $\#(w_1, w_2)$ with $w_1 \neq w_2 \in F^+ = p^f(p^f - 1) = |G|$.

Week 13 Hall subgroup

Def Hall subgroup

let G be a finite group of order $|G| = p_1^{f_1} p_2^{f_2} \dots p_t^{f_t}$ where p_i 's are the prime divisors of $|G|$. Let

$$\pi(G) (= \pi(|G|)) = \{p_1, \dots, p_t\}.$$

For a subset $\pi \subset \pi(G)$, a subgroup $H \leq G$ is called a Hall π -subgroup of G if $\pi(H) = \pi$ and $\gcd(|H|, |G|/|H|) = 1$.

① If $|\pi| = 1$, then a Hall π -subgroup is a Sylow subgroup.

② Hall subgroups do not necessarily exist

③ Hall π -subgroups are not necessarily conjugate.

Notation

$G_\pi \rightarrow$ Hall π -subgroup $\Rightarrow \pi$ in fact takes $\pi \cap \pi(G)$

$G_p \rightarrow$ if $\pi = \{p\}$.

eg. $G = A_5$. $|G| = 2^2 \cdot 3 \cdot 5$.

$\pi' \rightarrow \pi(G) \setminus \pi$.

$$G_{\{2,7\}} = 2^2 \cdot 7$$

$$G_{\{2,3\}} = A_4$$

$$G_5 = C_5 = G_{\{2,3\}}$$

example

① $G = A_5$. $|G| = 60$.

$\pi = \{2, 3\}$ A_4

$\{3, 5\}$ $\times$

$\{2, 5\}$ $\times$

pf: If H is a Hall $\{2, 5\}$ -subgroup. Then $[G:H]$ is of size 3, then $G \leq \text{Sym}([G:H]) \cong S_3$ which is impossible.

$\downarrow$ $G = A_5$ is simple, hence $G \rightarrow \text{Sym}([G:H])$ $\ker = 1$

Similarly for $\{3, 5\}$.

② $G = \text{PSL}_2(11)$ is simple. $|G| = 5 \cdot 11 \cdot 12$.

has subgroup H, K s.t. $H \cong A_4$, $K \cong D_{12}$ which are Hall $\{2, 3\}$ -subgroups of G . They are not isomorphic $\Rightarrow$ not conjugated.

Thm (Hall thm) solvable $\Leftrightarrow$ each Hall π -subgps exists.

Let G be a finite group with $|G| = p_1^{f_1} \dots p_t^{f_t}$ with $p_1, \dots, p_t$ being distinct primes. Then G is solvable $\Leftrightarrow$ G has Hall π -subgroups for each subset $\pi \subset \pi(G)$ pf:

When $\pi(G) = 1$, G is a p -group. Then G is solvable (nontrivial center + $N \triangleleft G$, N and G/N both solvable $\Rightarrow G$ solvable).

~~Assume inductively that Hall thm holds for gps of which the order of G has at most $t-1$ different prime divisors.~~
use induction on $|G|$.

$\Leftarrow$

If $|\pi(G)| = 2$, then G is solvable by Burnside's " $p^a q^b$ "-thm

Assume $t \geq 3$. Let $H_i = H_{p_i'}$ be a Hall $\{p_i'\}$ -subgp of G , for $1 \leq i \leq t$. Then H_i is solvable as $|\pi(H_i)| = t-1$

Let $p \parallel |H_i|$, i.e. $p = p_j$ for some $j \in \{2, \dots, t\}$. s.t. $O_p(H_i) \neq 1$.

(Let X be a group. Then $O_p(X)$ satisfies

① $O_p(X) \triangleleft X$

② $O_p(X)$ is a p -group.

③ $X/O_p(X)$ has no normal p -subgroups ($O_p(X/O_p(X)) = 1$).

$\Leftrightarrow O_p(X)$ is the intersection of all Sylow p -subgroups (otherwise full preimage is a larger normal p -subgroup)

(Such a p exists since a minimal normal subgroup of a solvable group is an elementary abelian p -group ($\mathbb{Z}_p^d$ for some p prime, d > 0) and must be contained in $O_p(H_i)$.)

Then p divides $|H_2|$ or $|H_3|$, say $p \mid |H_2|$. Let P be a Sylow p -subgroup of H_2 . Then P is a Sylow p -subgroup of G and $Op(H_1) \leq P^g \leq H_2^g$ for some g by Sylow thm.

Further, we have $G = H_1 H_2^g$
 $(H_1 H_2^g \leq G. \quad |H_1 H_2^g| = \frac{|H_1||H_2^g|}{|H_1 \cap H_2^g|} \geq \frac{(p_2^{f_2} p_3^{f_3} \dots p_t^{f_t})(p_1^{f_1} p_3^{f_3} \dots p_t^{f_t})}{p_3^{f_3} \dots p_t^{f_t}}$

Let $N = \langle Op(H_1)^x \mid x \in G \rangle$ the normal closure of $Op(H_1)$ in G . Then $N \triangleleft G$ and $N \leq H_2^g$ as $\forall x \in G, x = x_1 x_2$ for some $x_1 \in H_1, x_2 \in H_2^g$, then $Op(H_1)^x = Op(H_1)^{x_1 x_2} = Op(H_1)^{x_2} \leq H_2^g$. Thus N is a proper normal subgroup of G . For a Hall π -subgroup H of G , $H \cap N$ is a Hall π -subgroup of N and $H/H \cap N \cong HN/N$ is a Hall π -subgroup of G/N .
 $(|N|)_\pi = |H|_\pi \Rightarrow |N \cap H|_\pi = \frac{|H|_\pi |N|_\pi}{|HN|_\pi} = |N|_\pi$

By induction, N and G/N are both solvable, so G is solvable.
 $\Rightarrow$

Let G be solvable. Let $M \triangleleft_{\min} G$. Then $M = Z_p^d$ for some prime p and $d \geq 1$ and $\bar{G} = G/M$ has order properly dividing $|G|$. Then $\bar{G}$ has Hall ϕ -subgroups for all $\phi < \pi(\bar{G})$ (By $\bar{G}$ solvable and use induction).

Take a subset $\pi \subset \pi(\bar{G})$. Let $\bar{H}$ be a Hall π -subgroup of $\bar{G}$.

If $p \in \pi$, then the full image of $\bar{H}$ under $G \rightarrow \bar{G}$ is $M \cdot \bar{H}$ which is a Hall π -subgroup of G .

If $p \notin \pi$, aim to find a subgroup of order $|H|$ of G .

Now $(p, |\bar{H}|) = 1$. Let $X := M \cdot \bar{H}$ be the full preimage of $\bar{H}$ under $G \rightarrow \bar{G}$. If $X < G$ then by induction X has a

Hall π -subgroup H' which is isomorphic to $\bar{H}$ and is a π -Hall of X ($X = MH' = M \cdot \bar{H}$).

π -subgroup of G . then $\bar{G} = \bar{H}$

Assume $X = G$. Let $\bar{Q} = O_q(\bar{G}) \neq 1$ for some $q \in \pi(\bar{G})$.

Then the full preimage of $\bar{Q}$ under $G \rightarrow \bar{G}$ has the form $M:\bar{Q} = M:Q \triangleleft G$ for some Sylow q -subgroup $Q \cong \bar{Q}$ of $M:\bar{Q}$. (full preimage of a normal subgroup is a normal subgroup).

1) Suppose $C_M(Q) \neq 1$

Then $C_M(Q) = Z(M:Q)_P$

($C_M(Q) \in Z(M:Q)$ since M abelian, P is a characteristic

$P \times Q_1$

Sylow p -subgp

$\rightarrow$ Sylow q -subgp.

subgroup of $P \times Q_1 = Z(M:Q)$, $Z(M:Q)$ is a characteristic subgroup of $M:Q \triangleleft G$, so $P \triangleleft G$. As $P \leq M$ _{min}, $P = M$ and thus $M:Q = M \times Q$.)

Now Q is characteristic of $M \times Q = M:Q \triangleleft G$ we have

$Q \triangleleft G$. Let $\bar{K}$ be a π -subgp of G/Q . Then the full preimage of $\bar{K}$ under $G \rightarrow G/Q$ is a Hall π -subgroup of G (case $p \in \pi$).

2) Suppose $C_M(Q) = 1$.

Note that $N_M(Q) = C_M(Q)$.

(let $x \in M$ s.t. $x \in N_M(Q)$, then $xyx^{-1}y^{-1} \in M \cap Q = \{1\}$).

Claim: $N_G(Q) \cong N_{\bar{G}}(\bar{Q}) = \bar{G}$ (as $\bar{Q} \triangleleft \bar{G}$).

let $\bar{g} \in \bar{G} = N_{\bar{G}}(\bar{Q})$ and let g be a preimage of $\bar{g}$ in $G \rightarrow \bar{G}$.

then $\exists$ a unique $x \in M$ s.t. $gx \in N_G(Q)$

($\bar{g}^{-1}\bar{Q}\bar{g} = \bar{Q} \Rightarrow g^{-1}(M:Q)g = M:Q = M:g^{-1}Qg$)

i.e. $g^{-1}Qg$ is a Sylow q -subgroup of $M:Q$ so $\exists x \in M$

s.t. $x^{-1}(g^{-1}Qg)x = Q^{gx} = Q$ by Sylow thm. i.e. $gx \in N_G(Q)$.

Suppose $gx' \in N_G(Q)$ with $x' \in M$, then $x^{-1}x' = (gx)^{-1}gx' \in N_G(Q)$.

since $x^{-1}x' \in M$, $x^{-1}x' \in N_G(\bar{Q}) \cap M = N_M(\bar{Q}) \neq \{1\} \Rightarrow x = x'$

Define $\gamma: N_G(\bar{Q}) \rightarrow N_G(\bar{Q})$

$$\bar{g} \mapsto gX$$

Suppose $g_1x_1 = g_2x_2$ for some $x_1, x_2 \in M$.

Then $g_2^{-1}g_1 = x_2x_1^{-1} \in M$. so $g_2^{-1}g_1 = \overline{x_2x_1^{-1}} = 1$ so

$\bar{g}_2 = \bar{g}_1$ i.e. γ is injective.

so $|N_G(\bar{Q})| \leq |N_G(\bar{Q})|$.

Then "=" must hold and we are done ...

(if "<", then $MN_G(\bar{Q})/M \leq G/M = \bar{G}$ while

$MN_G(\bar{Q})/M = N_G(\bar{Q})/M \cap N_G(\bar{Q}) = N_G(\bar{Q})/N_M(\bar{Q}) = N_G(\bar{Q})$. contradiction). $\square$

Remark Moreover, for G solvable, Hall π -subgroups of G are conjugated for any $\pi \leq \pi(G)$.

pf: let G be solvable $\exists M \triangleleft G$ s.t. $|G/M| = p$ for some prime p

(let $N \triangleleft_{\min} G$. Then $\bar{G} = G/N$ has a subgroup $\bar{L}$ s.t. $|\bar{G}/\bar{L}| = p$ by induction. let L be the full preimage of $\bar{L}$ under $G \rightarrow \bar{G}$. Then $\frac{|G|}{|L|} = \frac{\frac{|G|}{|N|}}{\frac{|L|}{|N|}} = \frac{|\bar{G}|}{|\bar{L}|} = p$).

Let H, K be Hall π -subgroups of G . If $p \notin \pi$, then $H, K \leq M$. So H, K are conjugated by induction.

$$|H \cap M|_{\pi} = \frac{|H|_{\pi} |M|_{\pi}}{|M|_{\pi}} = |H|_{\pi}$$

Assume $p \in \pi$. Then $H/H \cap M \cong \mathbb{Z}_p \cong K/K \cap M$ and $H \cap M, K \cap M$ are Hall π -subgroups of M . By induction, $(H \cap M)^x = K \cap M$ for some $x \in M$. Now H^x, K are Hall π -subgroups of G . with $K = \langle K \cap M, g \rangle$ for some $g^p \in K \cap M$.

Date

Week 15 nilpotent groups

Thm Frattini argument

Let G be finite and $N \triangleleft G$ with

Then for $P \in \text{Syl}_p(N)$ we have $G = N_G(P)N$.

pf:

By Sylow thm, N is transitive on $\text{Syl}_p(N)$ by conjugation.

Since $N \triangleleft G$, G induces an action on $\text{Syl}_p(N)$.

$N \triangleleft G$, $\forall Q \in \text{Syl}_p(N)$, $Q^g \in \text{Syl}_p(N)$. As N is transitive

on $\text{Syl}_p(N)$, we have $G = N N_G(P)$
transitive $\searrow$ G_P

Nilpotent groups

① Let G be a p -group, then $Z(G) \neq 1$. $Z(G/Z(G)) \neq 1$ (if G not abelian). Let K be the full preimage of $Z(G/Z(G))$ under $G \rightarrow G/Z(G)$. Then $Z(G) \triangleleft K \triangleleft G$. Continue in this way we get a normal series.

② Let $G = P \times Q$ where P is a p -group, Q is a q -group, p, q distinct primes. Then $Z(G) \neq 1$ and $G/Z(G) = P \times Q$.

Def nilpotent group

A group G is called nilpotent if it has a chain of subgroups

$$G = K_1 > K_2 > K_3 > \dots > K_{s+1} = 1$$

such that $K_i \triangleleft G$ and $K_i/K_{i+1} \leq Z(G/K_{i+1})$.

(Such a series is called a central series).

example not nilpotent groups.

① $A_4 = C_2^2 : C_3$ only one normal subgroup C_2^2

$$\begin{array}{c} A_4 \\ \parallel \\ K_1 \end{array} > \begin{array}{c} C_2^2 \\ \parallel \\ K_2 \end{array} > \begin{array}{c} 1 \\ \parallel \\ K_3 \end{array}$$

$$C_2^2 = K_2 = K_2/K_3 \leq Z(A_4/K_3) = 1 \quad \text{KOKORO contradiction}$$

② S_4 has normal subgroups A_4, C_2^2 ,
Then $S_4 > A_4 > C_2^2 > 1$ not possible since $C_2^2 \not\leq Z(S_4)$.

③ D_{2p} $p \geq 3$ prime.

It has only one normal subgroup $\langle a \rangle$ if $D_{2p} = \langle a, b \mid a^p = b^2 = 1, a^b = a^{-1} \rangle$.

Then $D_{2p} > \langle a \rangle > 1$

But $\langle a \rangle \not\leq Z(D_{2p})$.

Thm equivalence definitions of being a nilpotent group.

(1) G is nilpotent

(2) If $H \leq G$, then $N_G(H) \not\leq H$.

(3) each maximal subgroup M of G is normal and is of prime index (i.e. $G/M \cong \mathbb{Z}_p$ for some p).

(4). G is a direct product of its Sylow subgroups i.e. each Sylow subgroup is normal.

Example

① Abelian groups are nilpotent groups.

② Subgroups and factor groups of nilpotent groups are nilpotent.

③ $G = A_4 = C_2^2 : C_3$. C_3 is maximal.

But it is not prime nor of prime index.

pf of thm :

(1) $\Rightarrow$ (2).

G has a chain of subgroups

$$G = K_1 > K_2 > \dots > K_{s+1} = 1$$

let $H \leq G$. Then $\exists i$ such that $K_{i+1} \leq H$ and $K_i \not\leq H$.

Then H/K_{i+1} and $Z(G/K_{i+1})$ centralizes each other. As

$K_i/K_{i+1} \leq Z(G/K_{i+1})$, for any $h \in H$, $y \in K_i$,

$$hyh^{-1}y^{-1} \in K_{i+1} \subset H$$

Hence $yh^{-1}y^{-1} \in h^{-1}H = H$.

i.e. $y \in N_G(H)$ thus $K_i \in N_G(H)$

Thus $N_G(H) \geq HK_i > H$ as in (2).

(2) $\Rightarrow$ (3). $\rightarrow$ notation of maximal group.

Let $M \subsetneq G$, then $M \neq N_G(M) = G$. So $M \triangleleft G$.

Then G/M does not have a nonidentity proper subgroup as M is maximal, so $|G/M|$ is prime as in (3).

(3) $\Rightarrow$ (4).

Let $P \in \text{Syl}_p(G)$. Then $P < G$ and $P < N_G(P) \leq G$.

Suppose $N_G(P) \neq G$. Then $\exists M$ s.t. $N_G(P) < M < G$.

Then $M \triangleleft G$ and by Frattini argument, $G = N_G(P)M = M$, contradiction. hence $P \triangleleft N_G(P) = G$ as in (4).

(4) $\Rightarrow$ (1).

Let $|G| = p_1^{e_1} p_2^{e_2} \dots p_t^{e_t}$ with $p_1, \dots, p_t$ be distinct primes. Let $P_i \in \text{Syl}_{p_i}(G)$. Then $P_i \triangleleft G$ and $G = P_1 \times \dots \times P_t$.

Let $L_1 = Z(G)$. Then $L_1 \neq 1$ and $\bar{G} = G/L_1$ is again a direct sum of its Sylow subgroups.

Let $\bar{L}_2 = Z(\bar{G})$. and let L_2 be the full preimage of $\bar{L}_2$ under $G \rightarrow \bar{G}$. Then $L_2 \triangleleft G$ as $\bar{L}_2 \triangleleft \bar{G}$ and $L_2/L_1 = Z(G/L_1)$.

Let $\bar{L}_3 = Z(G/L_2)$ and let $L_3 = L_2 \cdot \bar{L}_3$ be the full preimage of $\bar{L}_3$ under $G \rightarrow G/L_2$. Then $L_3 \triangleleft G$ and $L_3/L_2 = Z(G/L_2)$.

Repeating this process produce a chain of subgroups

$$1 < L_1 < L_2 < L_3 < \dots < L_s = G.$$

such that $L_i \triangleleft G$ and $L_i/L_{i-1} = Z(G/L_{i-1})$ as in (1). $\square$

Fratini subgroup & nilpotent group

Def Fratini subgroup.

Let G be a finite group. Then G must have many maximal subgroups say $\text{Max } G = \{M \triangleleft G\}$. Then the intersection of all maximal subgroups is called the Fratini subgroup of G , denoted by $\Phi(G)$.

Corollary (to be obtained)

- (1) $\Phi(G)$ is nilpotent
- (2) G nilpotent $\Leftrightarrow G/\Phi(G)$ nilpotent
- (3) (Wielandt) G nilpotent $\Leftrightarrow G' \leq \Phi(G)$.
- (4) If G is a p -group, then $G/\Phi(G)$ is elementary abelian.

Observation

- ① $\Phi(G)$ is a characteristic subgroup of G .
- ② If $N \triangleleft G$, then $\Phi(N) \triangleleft G$. (by ①).

Def non-generator

An elt x of G is called a non-generator if $\langle x, S \rangle = G$ then $\langle S \rangle = G$ for any $S \subset G$.

Example

- ① $G = \langle a \rangle = C_{p^2}$ Then each elt of $\langle a^p \rangle$ is a nongenerator
- ② Q_8 has nongenerators of order 1 or 2

use lemma below.

③ D_{2n}

④ A nonabelian simple group has no nonidentity nongenerator elt.
In fact, $\forall a, \exists b$ s.t. $\langle a, b \rangle = G$.

Lemma characterization of $\Phi(G)$.

$\Phi(G)$ consists of all non-generator of G .

pf:

$\subseteq$ let $x \in \Phi(G)$. Suppose $\langle x, S \rangle = G$ s.t. $\langle S \rangle \neq G$.

Then $\exists M < G$ s.t. $\langle S \rangle < M < G$. Then $x \in \Phi(G) \leq M$

so $G = \langle x, S \rangle \leq M$, not possible. Hence x is a nongenerator.

$\supseteq$ let $x \in G$ be a nongenerator. Suppose $M < G$ and $x \notin M$.

Then $\langle x, M \rangle = G$, not possible as $M < G$. So $x \in M$.

Thm (Gaschutz)

Let $N < G$ s.t. $N \leq \Phi(G)$. Then G/N is nilpotent.

$\Leftrightarrow G$ is nilpotent.

pf: It suffices to prove $\Rightarrow$

let $P \in \text{Syl}_p(G)$. We will show $P < G$.

Then $PN/N \in \text{Syl}_p(G/N)$. and $PN/N < G/N$. Hence $PN < G$. By Frattini argument $G = PN N_G(P) \stackrel{N \leq \Phi(G)}{=} PN_G(P) = N_G(P)$. So $P < G$. and thus G is nilpotent.

pf of corollary:

(1) It suffices to prove any Sylow p -subgp P of $\Phi(G)$ is normal in G .

By Frattini argument, $G = \Phi(G) N_G(P)$. If $N_G(P) \neq G$, $\exists M$ s.t. $N_G(P) < M < G$. But $\Phi(G) \leq M$, so $G \leq M$, contradiction. Hence $P \trianglelefteq G$.

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(2) Take $N = \Phi(G)$ in the thm.

(3) Suppose G is nilpotent. Then for any $M < G$, $M \trianglelefteq G$ and $G/M \cong C_p$ is abelian. Thus $G' \leq M$ and $\Phi(G) \geq G'$.

(4). By (3), $G/\Phi(G)$ is abelian. Moreover, $x^p \in \Phi(G) \forall x \in G$. Indeed, let $M < G$, if $x \in M$, we are done. If $x \notin M$, then since $G/M \cong C_p$, we have $x^p \in M$. Then $\forall \bar{g} \in G/\Phi(G)$, $\bar{g}^p = 1$. Hence $G/\Phi(G)$ is elementary abelian.

example

Let $G = C_{p^2} : C_q = \langle a \rangle : \langle b \rangle$, where $p \neq q$ primes.

Then $a^p b = b a^p \Leftrightarrow ab = ba \Leftrightarrow G/\langle a^p \rangle$ is abelian.

In particular, G is abelian $\Leftrightarrow G/\langle a^p \rangle \cong C_p : C_q$ abelian.

Def Fitting subgroup

The product of all normal nilpotent subgroups of G is called the Fitting subgroup of G , denoted by $F(G)$.
 $(\Rightarrow \Phi(G) < F(G))$.

Thm

Let G be solvable, and $F = F(G)$. $|G| = p_1^{e_1} p_2^{e_2} \dots p_t^{e_t}$. Then

(1) $F = O_{p_1}(G) \times \dots \times O_{p_t}(G)$ ($O_{p_1}(G) \times \dots \times O_{p_t}(G)$ characteristic subgp of

(2) $C_G(F) \leq F$ i.e. F is self-centralizing. (自中心化).

$(\Rightarrow G/C_G(F) \leq \text{Aut}(F))$.

Recall $O_p(G)$.

Let $p \mid |G|$. Then

$$O_p(G) := \begin{cases} \text{the largest normal } p\text{-subgp of } G \\ \text{the product of all normal } p\text{-subgroups of } G \\ \bigcap_{P \in \text{Syl}_p(G)} P \end{cases}$$

① $O_p(G)$ char G ($O_p(G)$ is a characteristic subgroup) and $\bar{G} := G/O_p(G)$ is s.t. $O_p(\bar{G}) = 1$.

② If $p \neq q$, then $\langle O_p(G), O_q(G) \rangle = O_p(G) \times O_q(G)$ char?

pf of thm:

(1) Let N be a normal nilpotent subgroup of G .

Then $N = Q_1 \times Q_2 \times \dots \times Q_t$ where $Q_i \in \text{Syl}_{p_i}(N)$.

Now Q_i char $N \triangleleft G$, so $Q_i \triangleleft G$ and thus

$Q_i \leq O_{p_i}(G)$. Hence $N \leq O_{p_1}(G) \times \dots \times O_{p_t}(G)$.

Since $O_{p_1}(G) \times \dots \times O_{p_t}(G)$ is a normal nilpotent subgroup of G , we have $F = O_{p_1}(G) \times \dots \times O_{p_t}(G)$.

(2). Let $C = C_G(F)$. Then $C \cap F \leq Z(F)$. Since F char G , so are C , CF and $C \cap F$. Let $W = C \cap F$.

$\bar{G} := G/W$. Suppose $\bar{C} \neq 1$. Then there exists p s.t.

$O_p(\bar{C}) \neq 1$ and $O_p(\bar{C})$ char $\bar{C}$ char $\bar{G}$. Then the full preimage of $O_p(\bar{C})$ $y := W$. $O_p(\bar{C}) \leq C$ char G .

Let $P \in \text{Syl}_p(y)$. Then $W_p \leq P \leq C$. By def of W and C , P centralizes W_p , so $y = W_p' \times P$.

Then P char y char G and $P \triangleleft G$ so that

$W_p < P \leq F$, a contradiction. Thus $O_p(\bar{C}) = 1$ and $C \leq Z(F) \leq F$. as $P \leq C \cap F$.

example

Let G be such that each Sylow subgroup is cyclic. Then G is solvable. Further, G is metacyclic, i.e. $\exists C \triangleleft G$ is cyclic and G/C is cyclic.

pf: Let $F = F(G)$. Then F is cyclic and $G/C_G(F) \leq \text{Aut}(F)$ is abelian. Since F is abelian, $C_G(F) = F$ and G/F is abelian. Since Sylow subgroups of G/F are cyclic, G/F is cyclic. Hence G is solvable.

(HW) ① G is as in example above.

show G has Hall subgroups N, H s.t. $G = N:H$ where each member of π is larger than members of π' .

② Let G be solvable and $|G| = 4m$ where m is odd and square-free. Characterize G .

③ example before

④ solvable

⑤ ①

$$\forall p \in \pi$$

$$|F|_p = |G|_p$$

other p go to π'